Enhancing Visa Processing Efficiency through Predictive Modeling

Contents

1. **About Data and Its Dictionary.....3**
2. **Exploratory Data Analysis 7**
    a. 2.1 Univariate Analysis
    b. 2.2 Bivariate Analysis
3. **Data Preprocessing…….…….. 11**
    a. 3.1 Outlier Detection and Treatment
    b. 3.2 Train-test & Validation split
4. **Model Building Original Data……. 12**
    a. Decision Tree
    b. Bagging
    c. Random Forest
    d. Ada-Boosting
    e. XGBoosting
5. **Model Building Oversample Data ……..26**
    a. Decision Tree
    b. Bagging
    c. Random Forest
    d. Ada-Boosting
    e. XGBoosting
6. **Model Building Undersample Data**……..**30**
    a. Decision Tree
    b. Bagging
    c. Random Forest
    d. Ada-Boosting
    e. XGBoosting
7. **Model Performance Improvement using Hyperparameter Tuning…….……35**
    a. Decision Tree
    b. Bagging
    c. Random Forest
    d. Ada-Boosting
    e. XGBoosting
8. **Model Performance Comparison and Final Model Selection……. 50**
9. **Actionable Insights & Recommendations…… 54**

About Data and its Dictionary

Dataset:

EasyVisa

Data Overview:

The data contains the different attributes of the employee and the employer.

Data Dictionary:

- case_id: ID of each visa application
- continent: Information of continent the employee
- education_of_employee: Information of education of the employee
- has_job_experience: Does the employee have any job experience? Y= Yes; N = No
- requires_job_training: Does the employee require any job training? Y = Yes; N = No
- no_of_employees: Number of employees in the employer's company
- yr_of_estab: Year in which the employer's company was established
- region_of_employment: Information of foreign worker's intended region of employment in the US.
- prevailing_wage: Average wage paid to similarly employed workers in a specific occupation in the area of intended employment. The purpose of the prevailing wage is to ensure that the foreign worker is not underpaid compared to other workers offering the same or similar service in the same area of employment.
- unit_of_wage: Unit of prevailing wage. Values include Hourly, Weekly, Monthly, and Yearly.
- full_time_position: Is the position of work full-time? Y = Full-Time Position; N = Part-Time Position

- case_status: Flag indicating if the Visa was certified or denied

Number of rows and columns:

```
The number of rows in data: 25480
The number of columns in data: 12
```

First Five Rows:

	case_id	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment
0	EZYV01	Asia	High School	N	N	14513	2007	West
1	EZYV02	Asia	Master's	Y	N	2412	2002	Northeast
2	EZYV03	Asia	Bachelor's	N	Y	44444	2008	West
3	EZYV04	Asia	Bachelor's	N	N	98	1897	West
4	EZYV05	Africa	Master's	Y	N	1082	2005	South

prevailing_wage	unit_of_wage	full_time_position	case_status
592.2029	Hour	Y	Denied
83425.6500	Year	Y	Certified
122996.8600	Year	Y	Denied
83434.0300	Year	Y	Denied
149907.3900	Year	Y	Certified

DataFrame Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25480 entries, 0 to 25479
Data columns (total 12 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   case_id                25480 non-null  object
 1   continent              25480 non-null  object
 2   education_of_employee  25480 non-null  object
 3   has_job_experience     25480 non-null  object
 4   requires_job_training  25480 non-null  object
 5   no_of_employees        25480 non-null  int64
 6   yr_of_estab            25480 non-null  int64
 7   region_of_employment   25480 non-null  object
 8   prevailing_wage        25480 non-null  float64
 9   unit_of_wage           25480 non-null  object
 10  full_time_position     25480 non-null  object
 11  case_status            25480 non-null  object
dtypes: float64(1), int64(2), object(9)
memory usage: 2.3+ MB
```

Checking Duplicate values:

```
0
```

Case ID column:

```
<bound method IndexOpsMixin.nunique of 0          EZYV01
1          EZYV02
2          EZYV03
3          EZYV04
4          EZYV05
          ...
25475    EZYV25476
25476    EZYV25477
25477    EZYV25478
25478    EZYV25479
25479    EZYV25480
Name: case_id, Length: 25480, dtype: object>
```

- **It is the unique identifier of each person, so we can drop this column.**

Checking Null values:

```
case_id                  0
continent                0
education_of_employee    0
has_job_experience       0
requires_job_training    0
no_of_employees          0
yr_of_estab              0
region_of_employment     0
prevailing_wage          0
unit_of_wage             0
full_time_position       0
case_status              0
dtype: int64
```

Target Column:

```
case_status
Certified    66.789639
Denied       33.210361
Name: proportion, dtype: float64
```

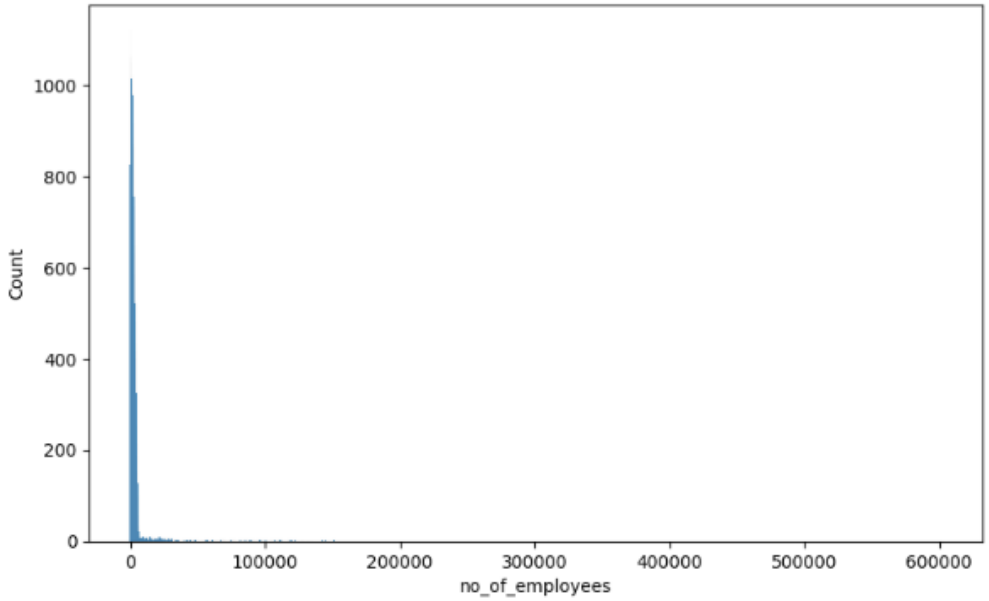
Statistical Information:

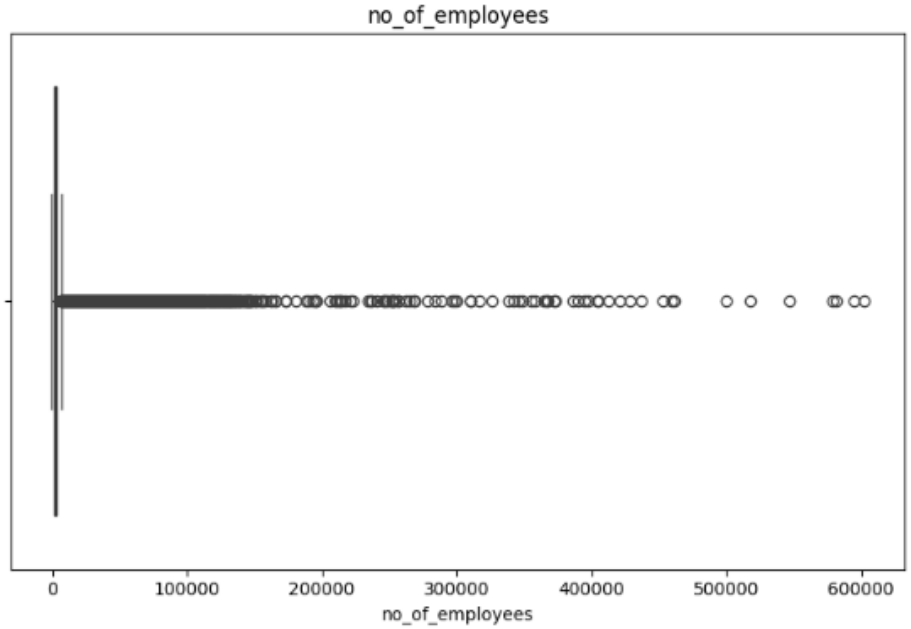
	count	unique	top	freq	mean	std	min	25%	50%	75%	max
continent	25480	6	Asia	16861	NaN	NaN	NaN	NaN	NaN	NaN	NaN
education_of_employee	25480	4	Bachelor's	10234	NaN	NaN	NaN	NaN	NaN	NaN	NaN
has_job_experience	25480	2	Y	14802	NaN	NaN	NaN	NaN	NaN	NaN	NaN
requires_job_training	25480	2	N	22525	NaN	NaN	NaN	NaN	NaN	NaN	NaN
no_of_employees	25480.0	NaN	NaN	NaN	5667.04321	22877.928848	-26.0	1022.0	2109.0	3504.0	602069.0
yr_of_estab	25480.0	NaN	NaN	NaN	1979.409929	42.366929	1800.0	1976.0	1997.0	2005.0	2016.0
region_of_employment	25480	5	Northeast	7195	NaN	NaN	NaN	NaN	NaN	NaN	NaN
prevailing_wage	25480.0	NaN	NaN	NaN	74455.814592	52815.942327	2.1367	34015.48	70308.21	107735.5125	319210.27
unit_of_wage	25480	4	Year	22962	NaN	NaN	NaN	NaN	NaN	NaN	NaN
full_time_position	25480	2	Y	22773	NaN	NaN	NaN	NaN	NaN	NaN	NaN
case_status	25480	2	Certified	17018	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Exploratory Data Analysis

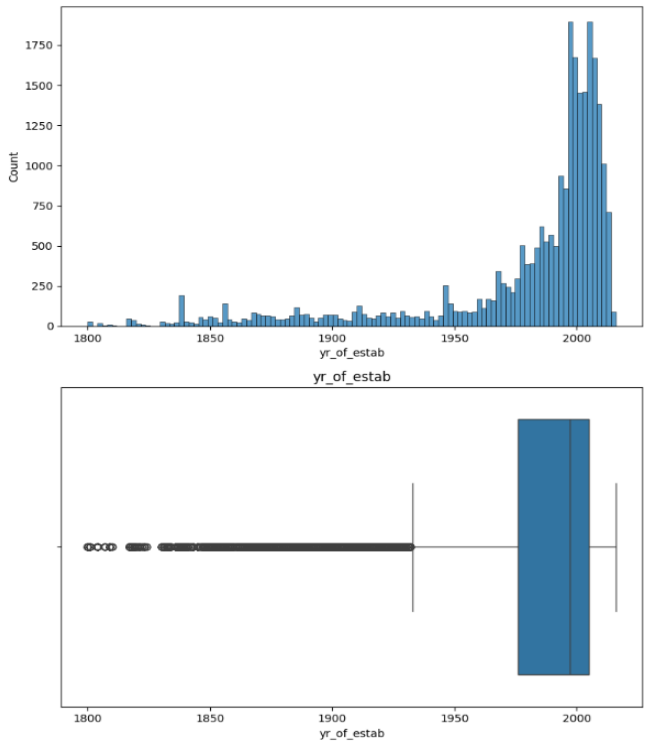
Univariate Analysis

No of Employees:

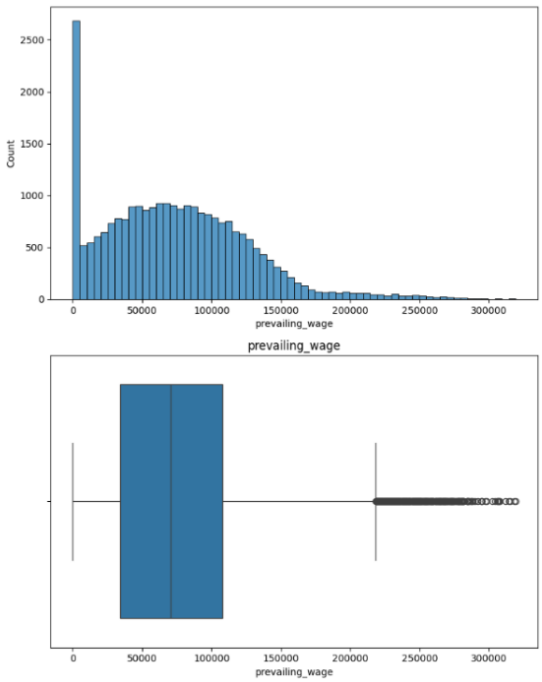




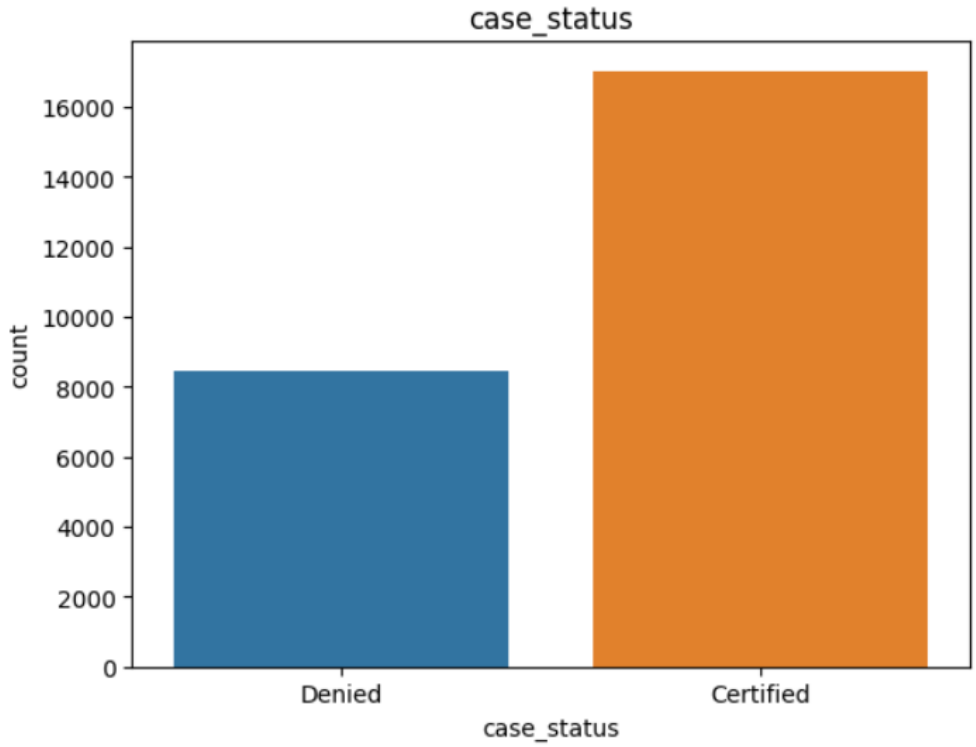
Year of Establish:



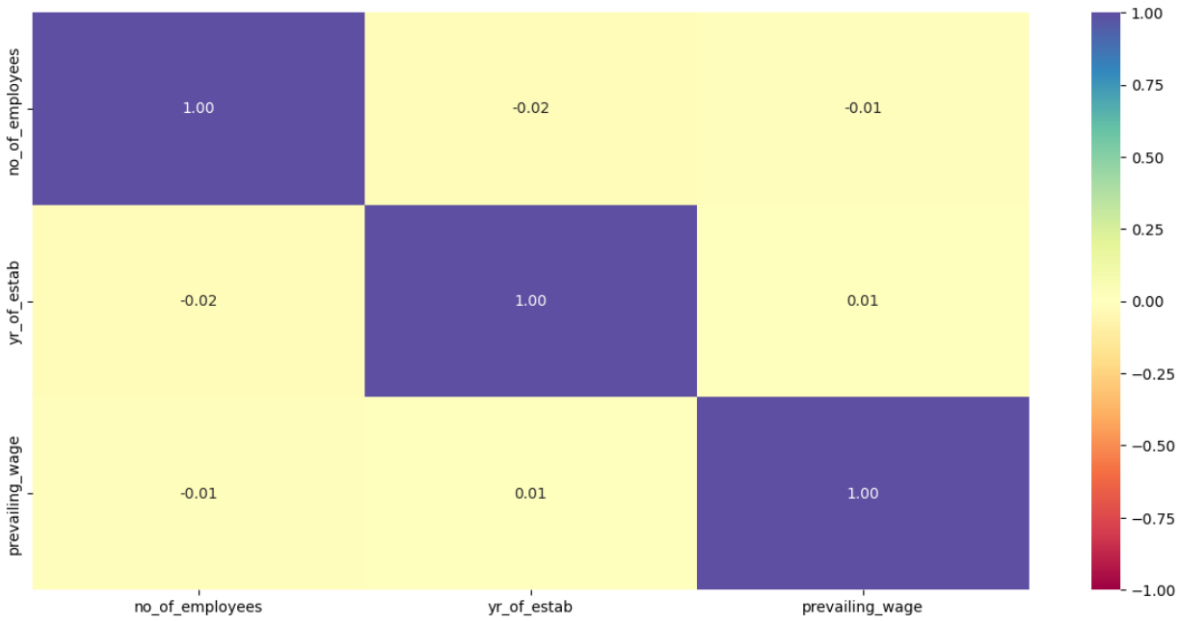
prevailing wage:



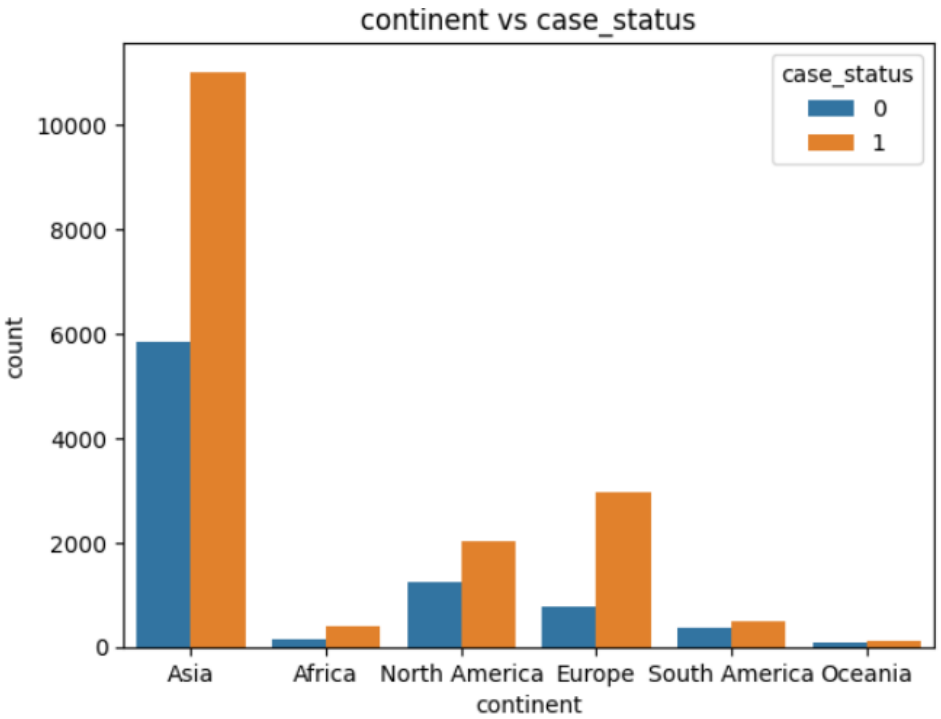
Case Status:



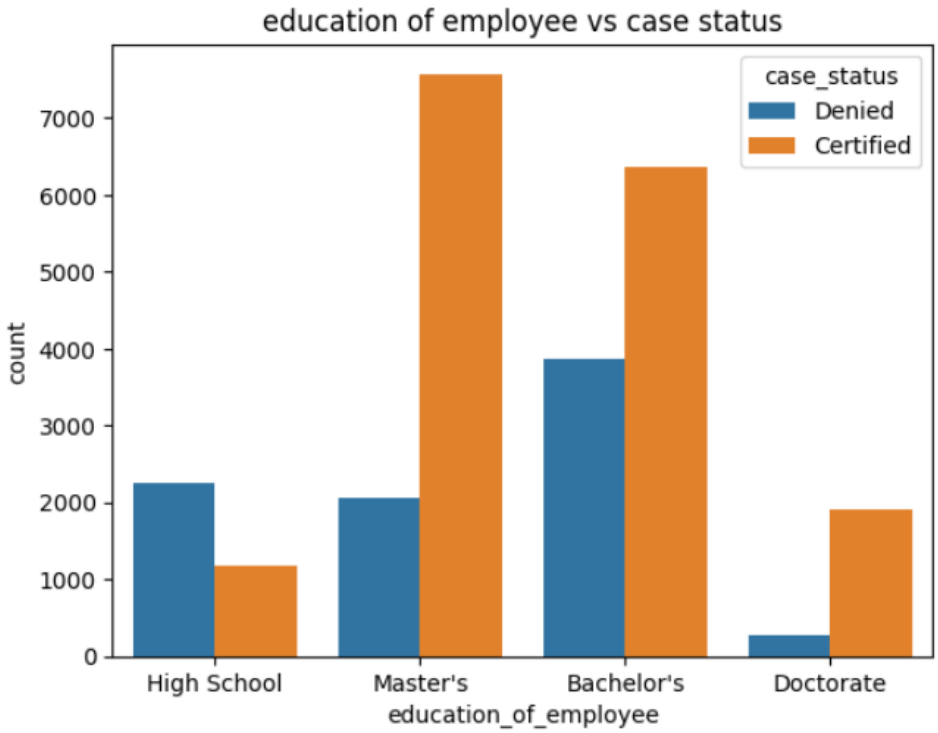
Bivariate Analysis



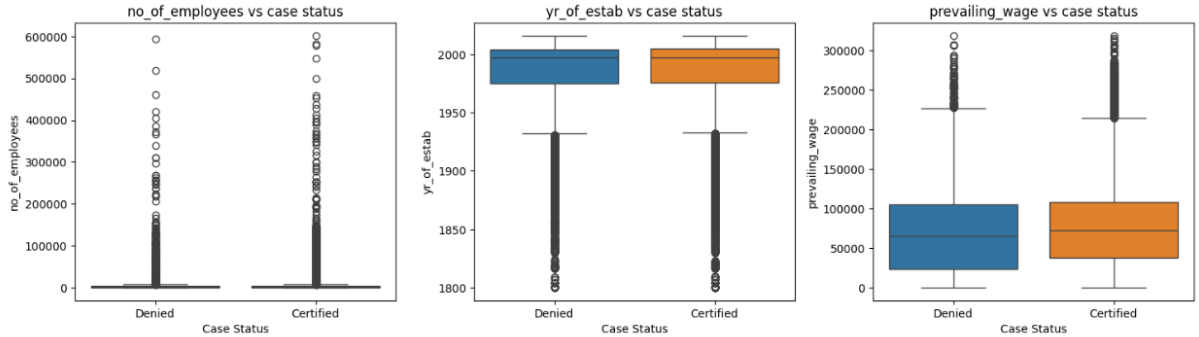
Continent vs Case Status:



Education of employee vs Case status:

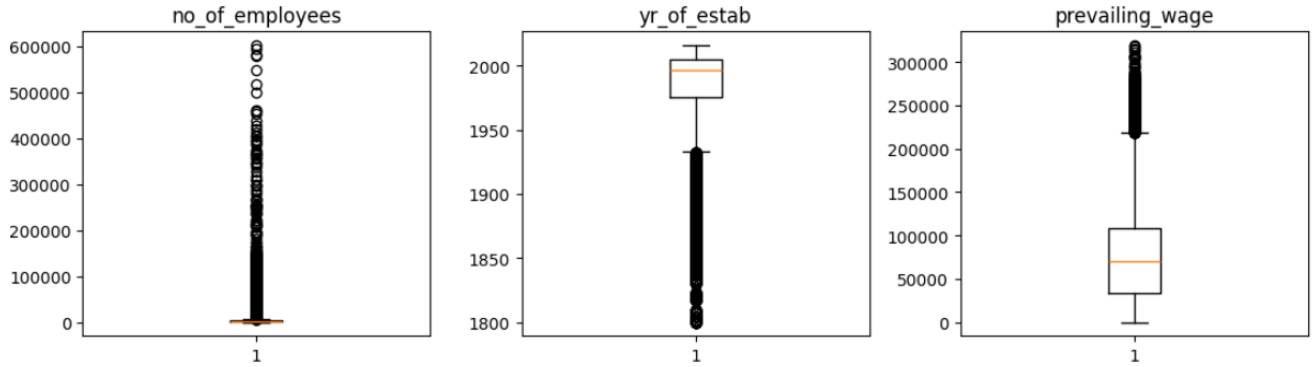


Case status vs Numerical Columns:

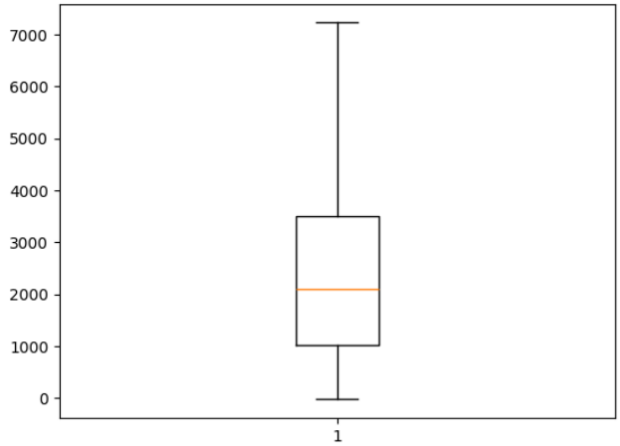


Data Preprocessing

Outlier Detection and Treatment



Year of establishment and prevailing wages are important, but there are more outliers in the number of employees, so we only treat the number of employees.



- The outliers have been treated.

Train, Test & Validation Split

```
Number of rows & columns in train data = (12485, 21)
Number of rows & columns in validation data = (5351, 21)
Number of rows & columns in test data = (7644, 21)
```

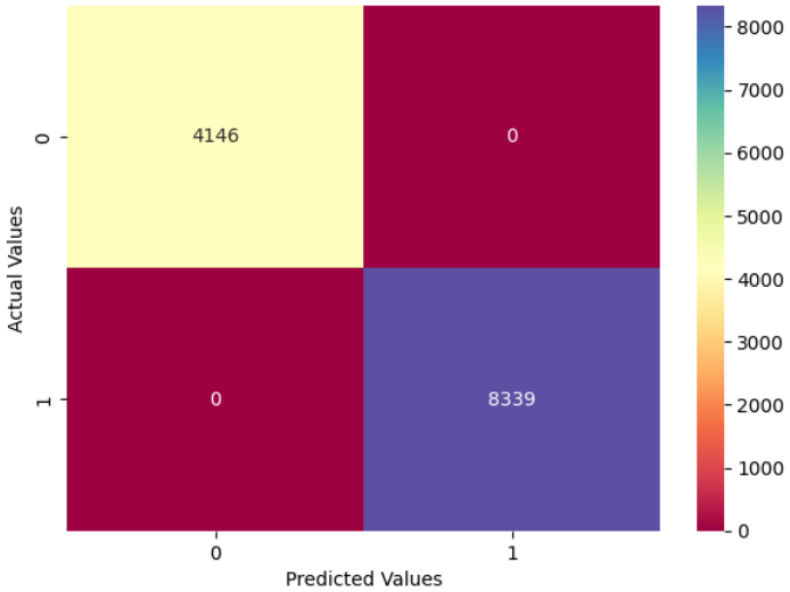
Scaled Data

```
Number of rows & columns in train data = (12485, 21)
Number of rows & columns in validation data = (5351, 21)
Number of rows & columns in test data = (7644, 21)
```

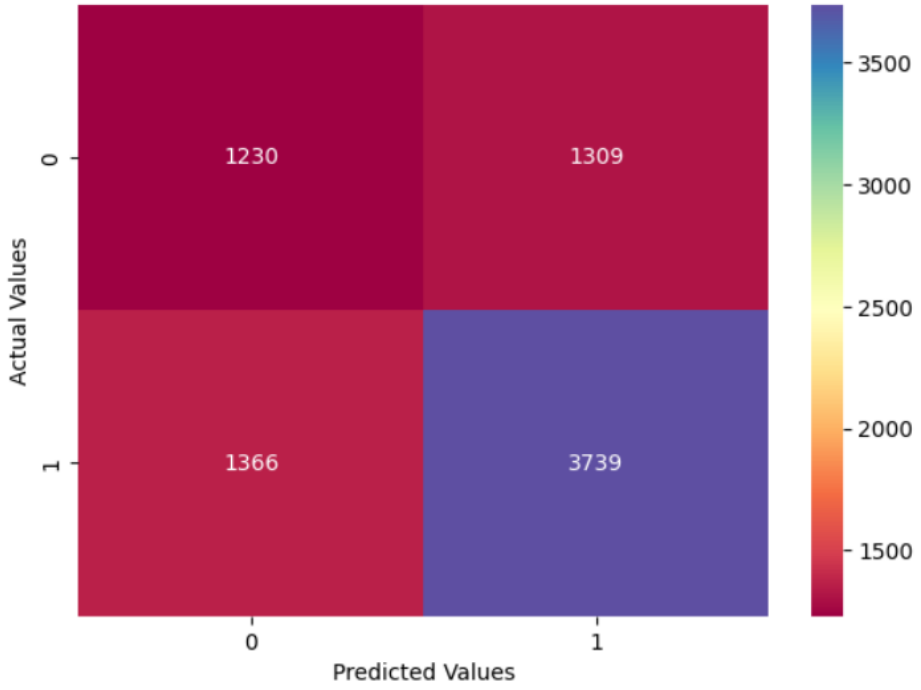
Model Building – Original Data

Decision Tree

Confusion Matrix (Train):



Confusion Matrix (Test):



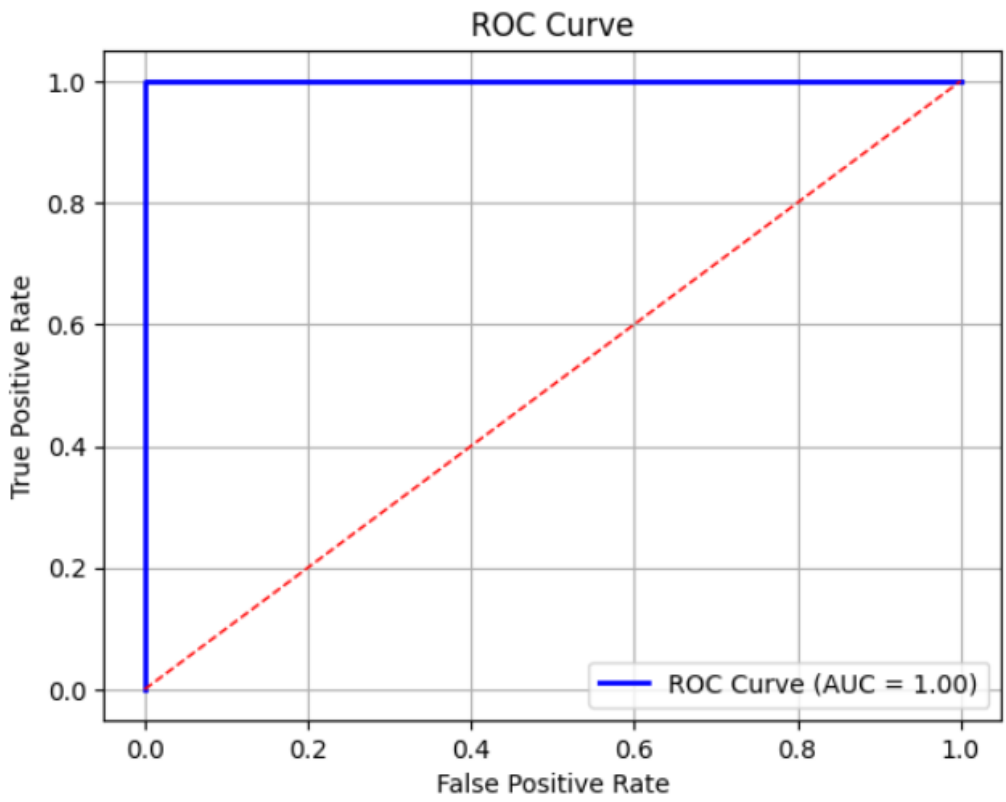
- True Positives (TP): The application **was certified** (approved) and the model correctly predicted it as **certified**.
- True Negatives (TN): The application **was denied** and the model correctly predicted it as **denied**.
- False Positives (FP): The model predicted the application as **certified,** but it was **denied**.
- False Negatives (FN): The model predicted the application as **denied** but it was **certified**.

Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

ROC – AUC Curve (Train):

```
AUC: 1.00
```



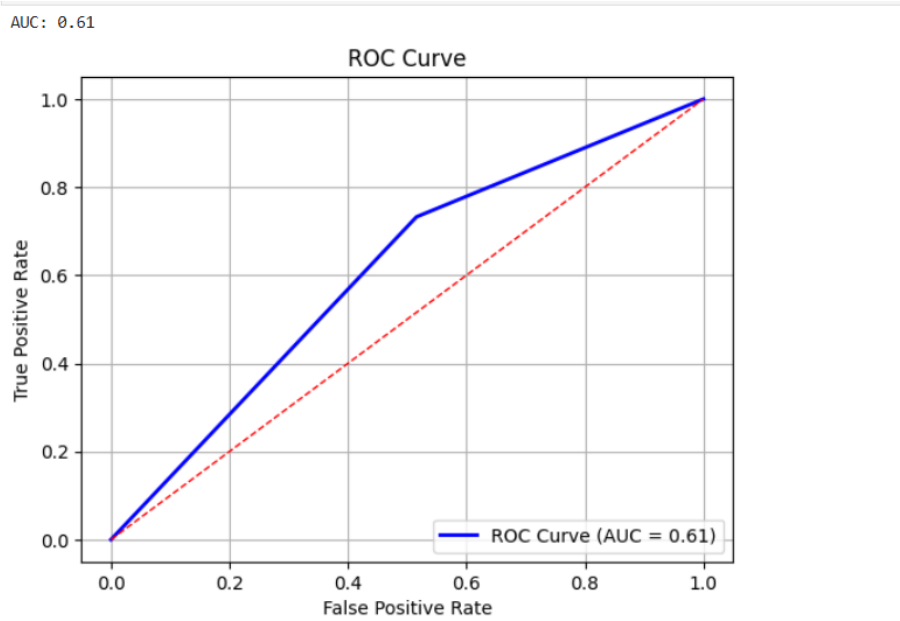
Insights:

- Perfect scores across all metrics, likely overfitting the training data.
- Recall = 1.0%
- It is perfect, and overfitting.

Performance Metrics (Test):

	Accuracy	Recall	Precision	F1
0	0.650052	0.732419	0.740689	0.736531

ROC – AUC Curve (Test):

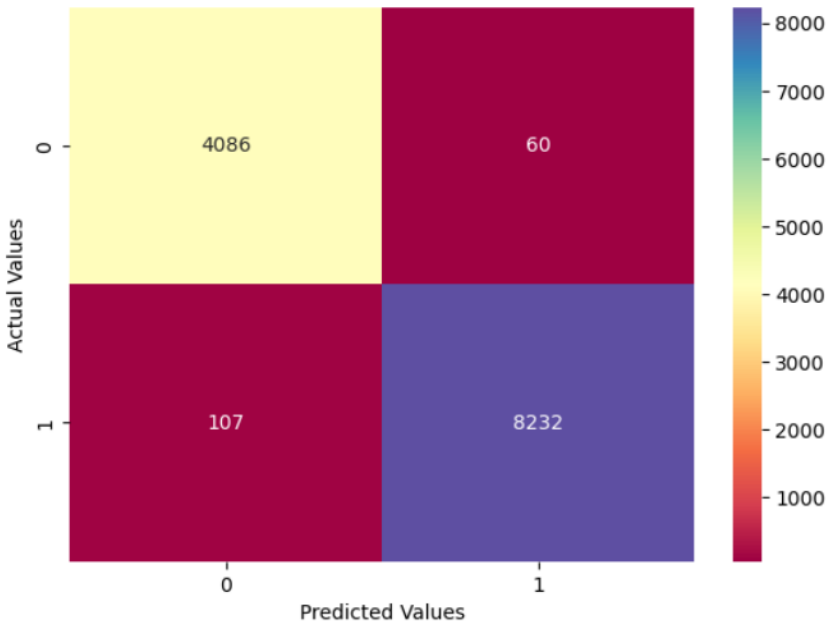


Insights:

- Indicates overfitting during training and poor generalization.
- Lowest performance across all metrics.

Bagging Model

Confusion Matrix (Train):

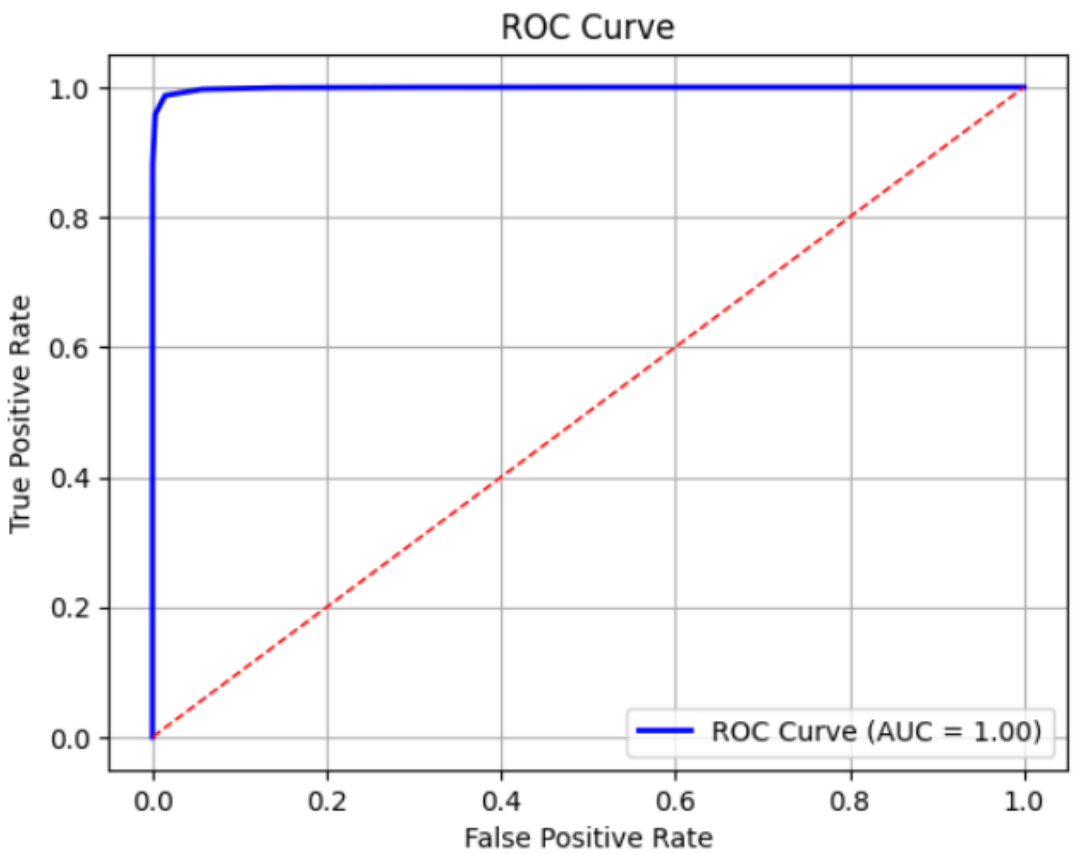


Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.986624	0.987169	0.992764	0.989959

ROC – AUC Curve (Train):

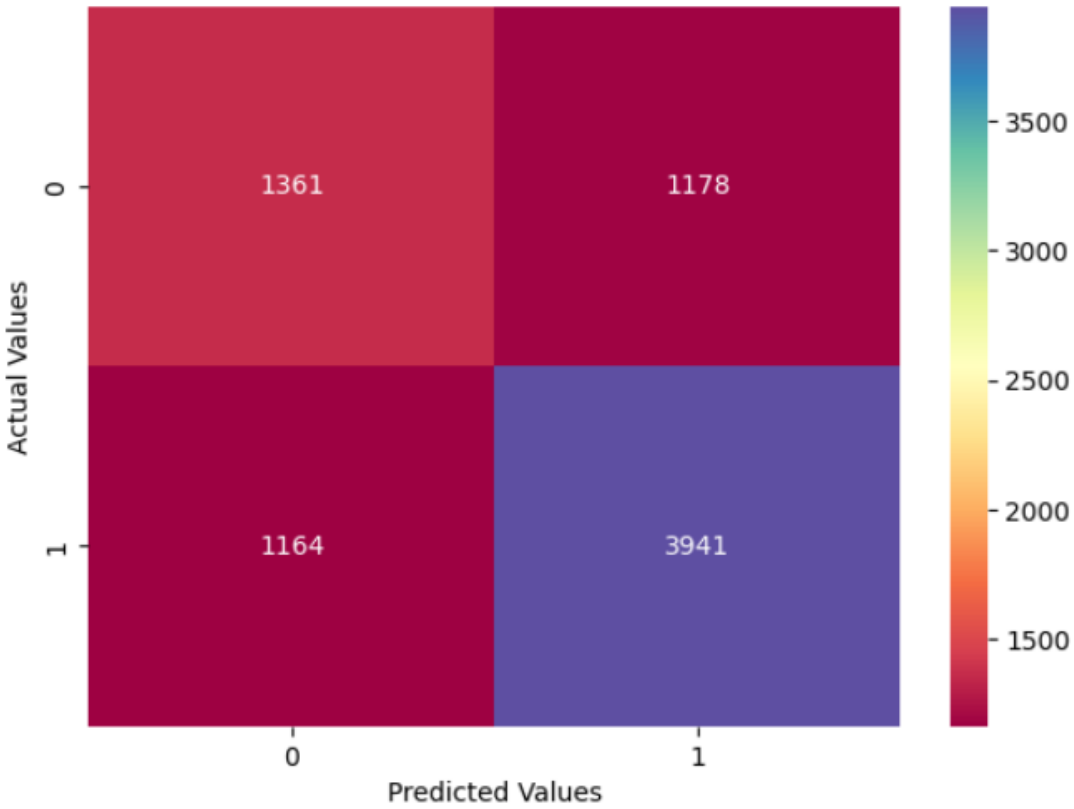
AUC: 1.00



Insights:

- Near-perfect scores.
- Classic sign of overfitting.

Confusion Matrix (Test):



Performance Metrics (Test):

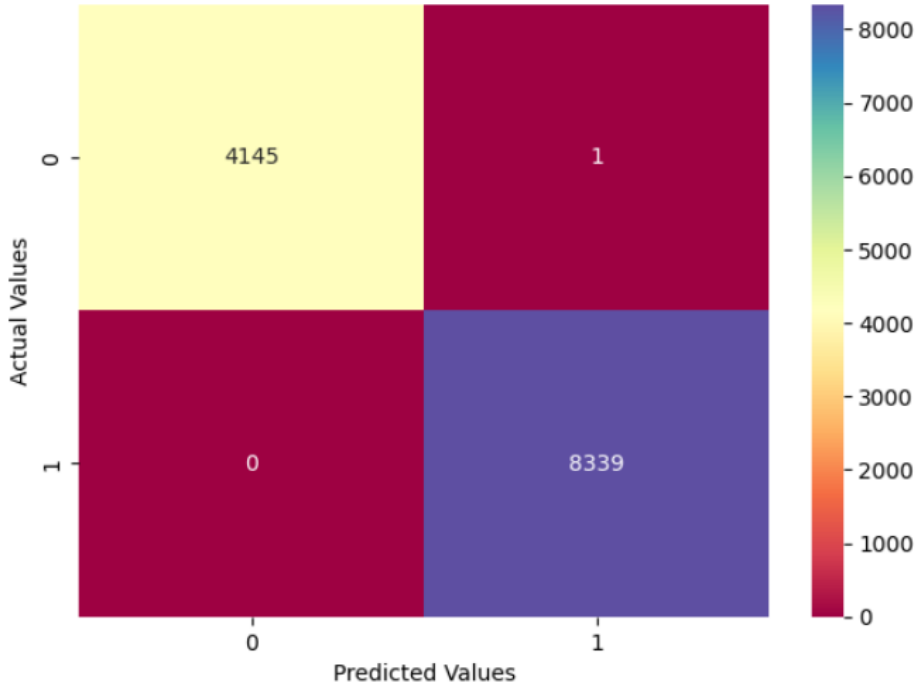
	Accuracy	Recall	Precision	F1
0	0.693616	0.771988	0.769877	0.770931

Insights:

- Moderate performance across all metrics.
- Balanced but not leading in any metric.

Random Forest Model

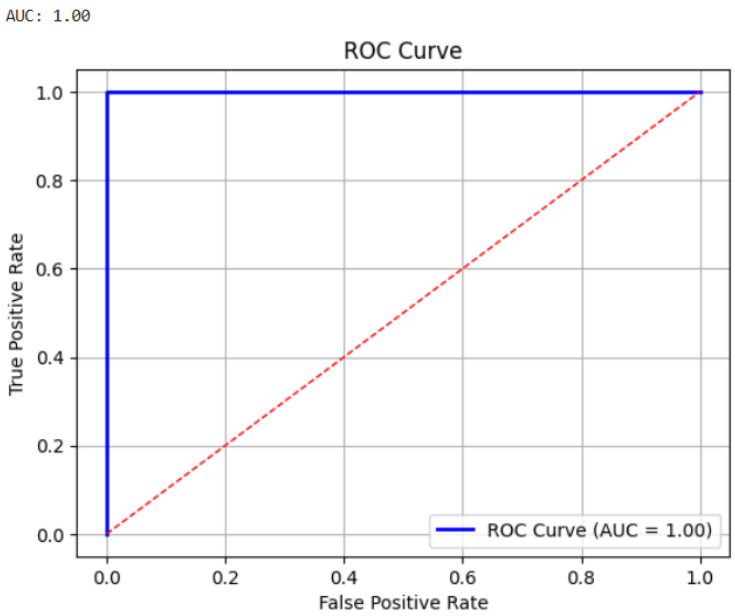
Confusion Matrix (Train):



Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.99992	1.0	0.99988	0.99994

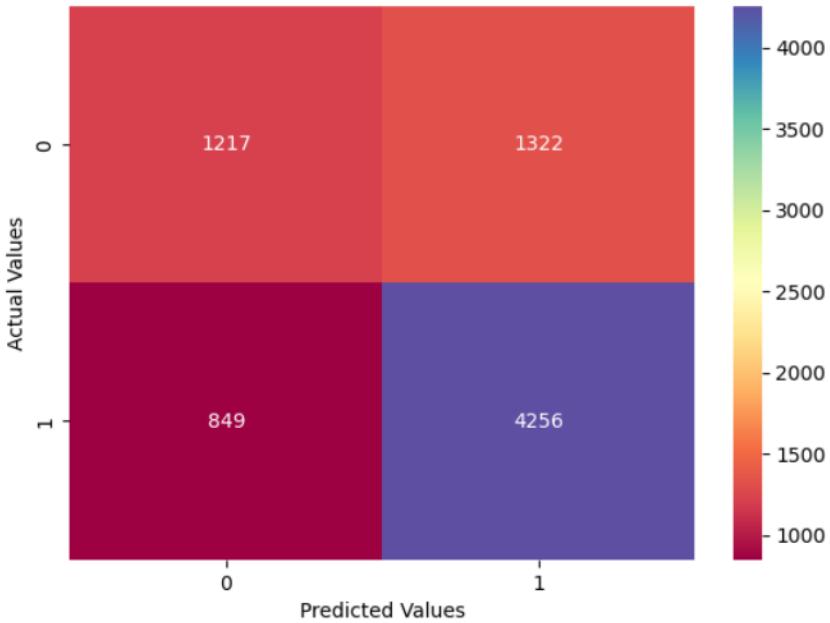
ROC – AUC Curve (Train):



Insights:

- Near-perfect performance, like Bagging.
- Excellent balance of precision and recall.

Confusion Matrix (Test):



Performance Metrics (Test):

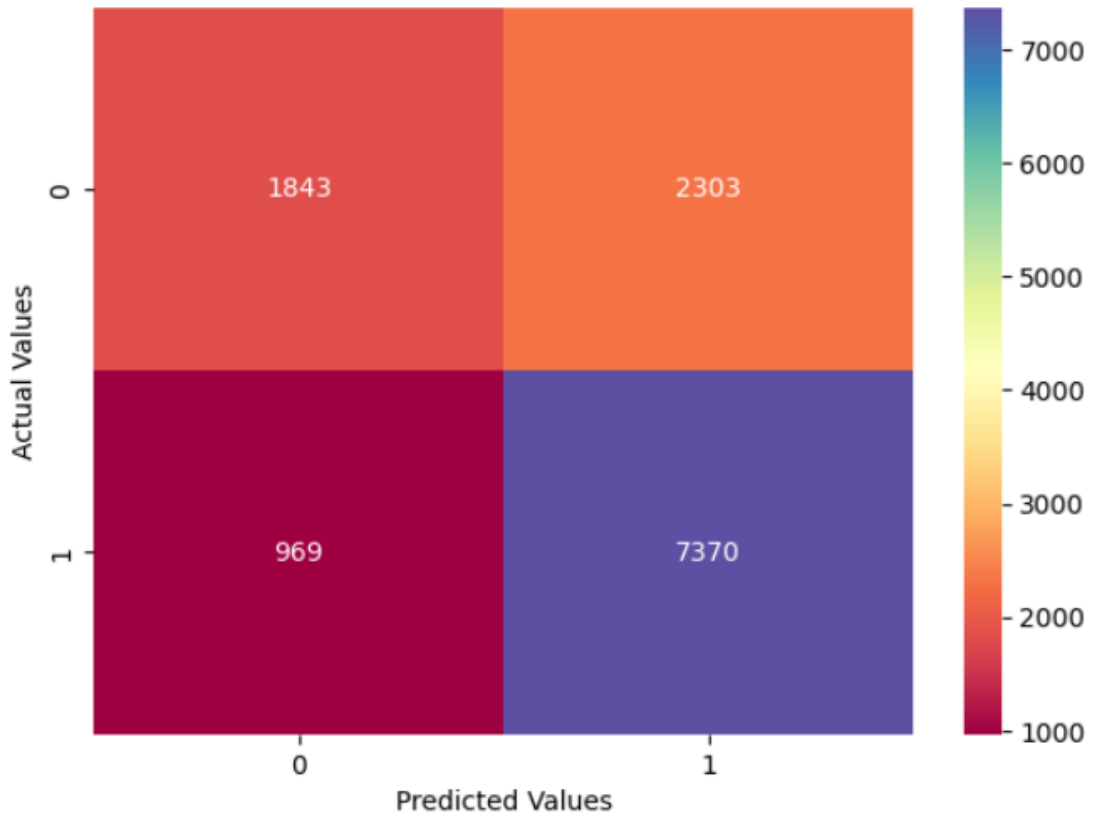
	Accuracy	Recall	Precision	F1
0	0.715986	0.833692	0.762997	0.79678

Insights:

- Strong recall and F1, but slightly lower accuracy
- Sign of Overfitting.

AdaBoost Model

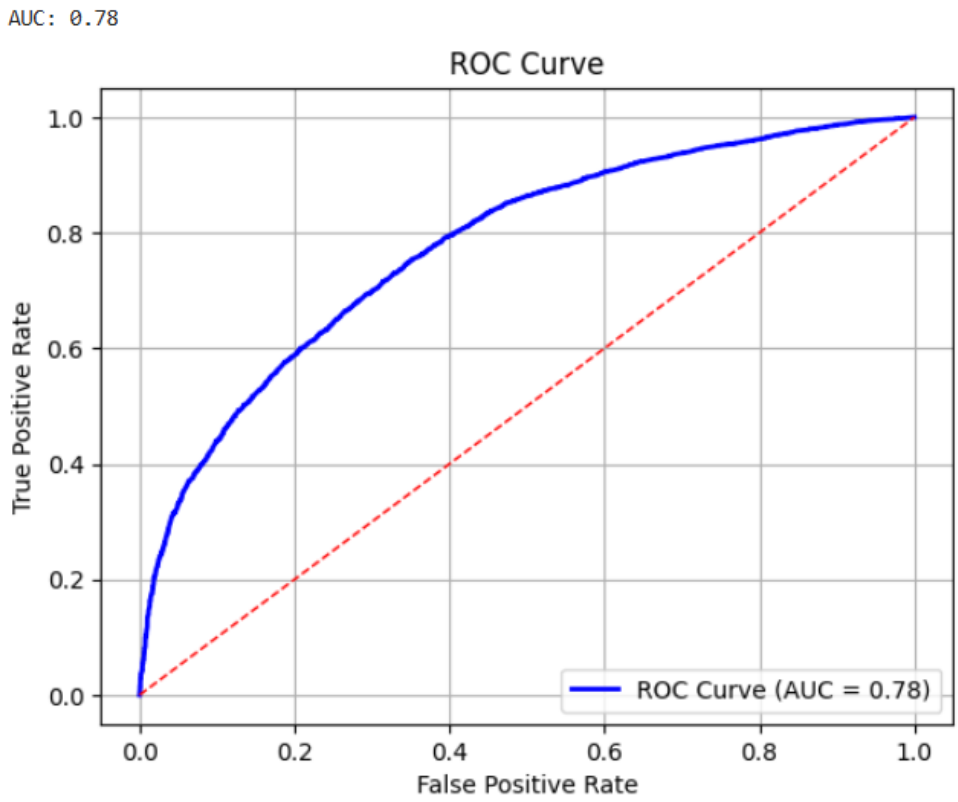
Confusion Matrix (Train):



Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.737926	0.883799	0.761915	0.818343

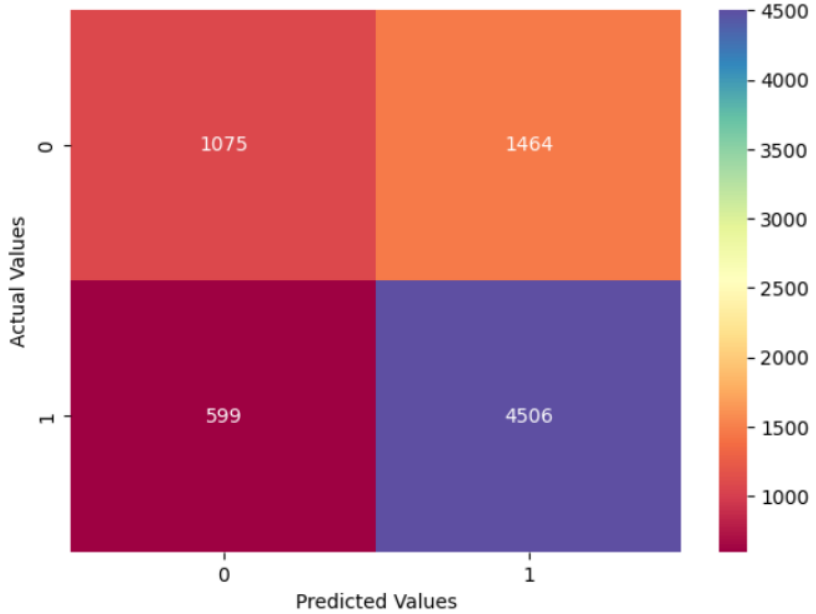
ROC – AUC Curve (Train):



Insights:

- Much lower training accuracy (0.738) compared to ensemble methods.
- Indicates less overfitting but may underfit if not tuned.

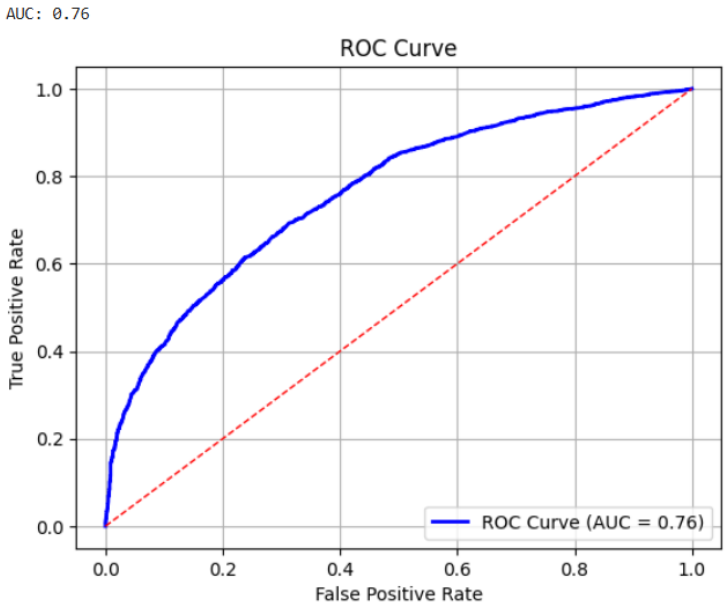
Confusion Matrix (Test):



Performance Metrics (Test):

	Accuracy	Recall	Precision	F1
0	0.730115	0.882664	0.754774	0.813725

ROC – AUC Curve (Test):

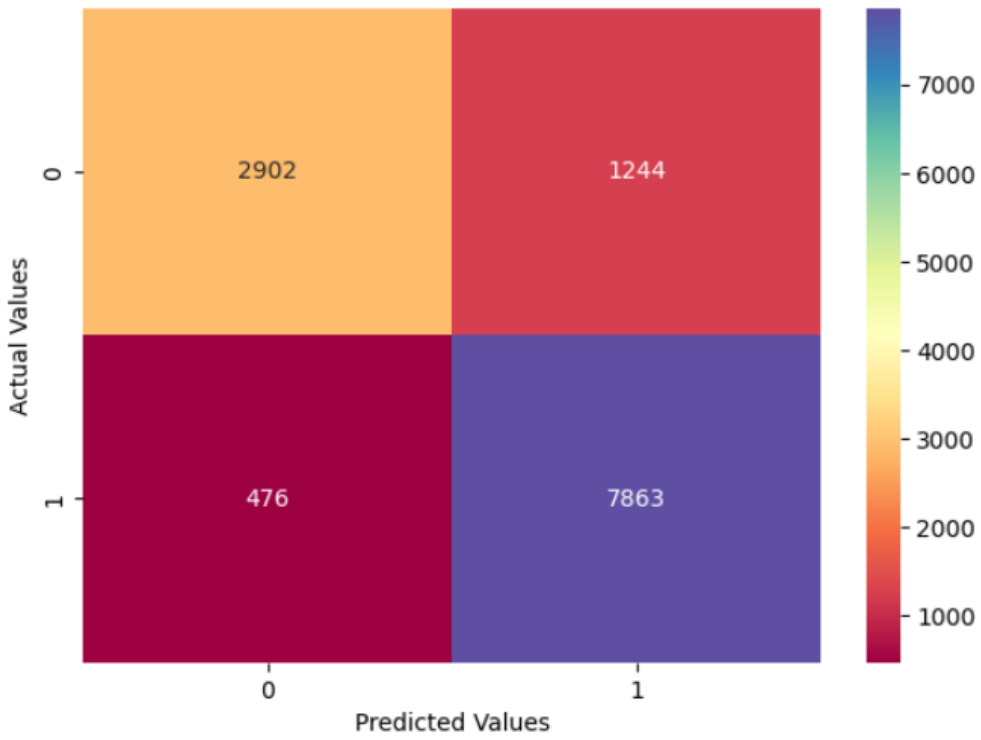


Insights:

- Best overall performer on test data.
- Highest accuracy, recall, and F1 score.
- Slightly lower precision, indicating more false positives, but excellent at identifying true positives.

XGBoost Model

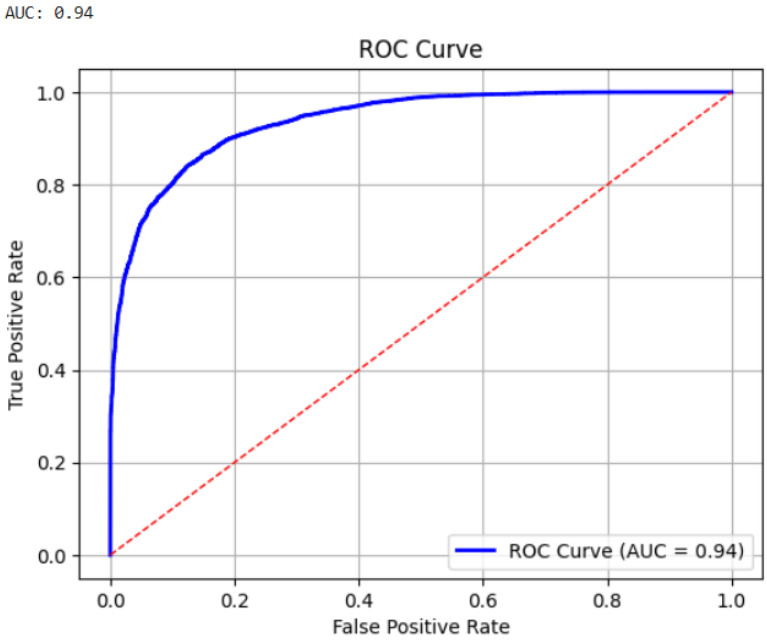
Confusion Matrix (Train):



Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.862235	0.942919	0.863402	0.90141

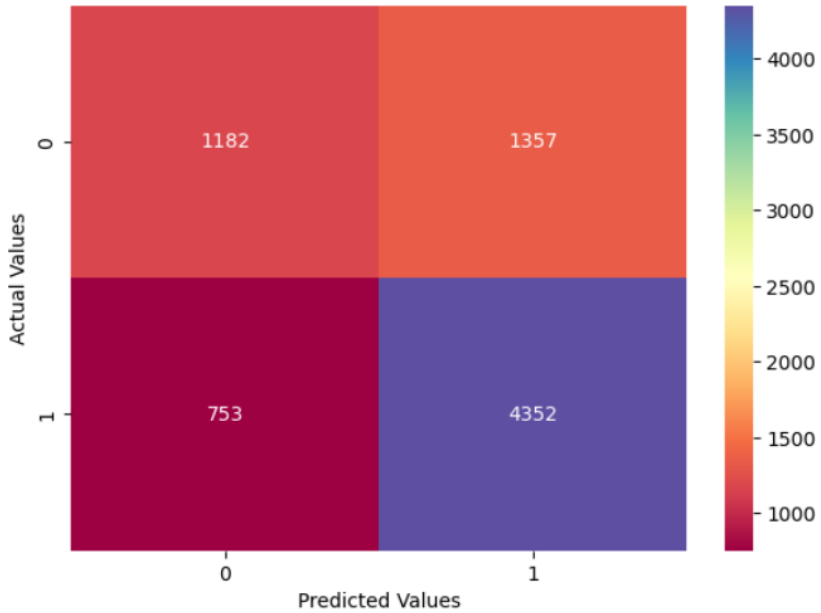
ROC – AUC Curve (Train):



Insights:

- Strong performance, especially in recall (0.943).
- Slightly lower precision (0.863), suggesting more false positives.

Confusion Matrix (Test):

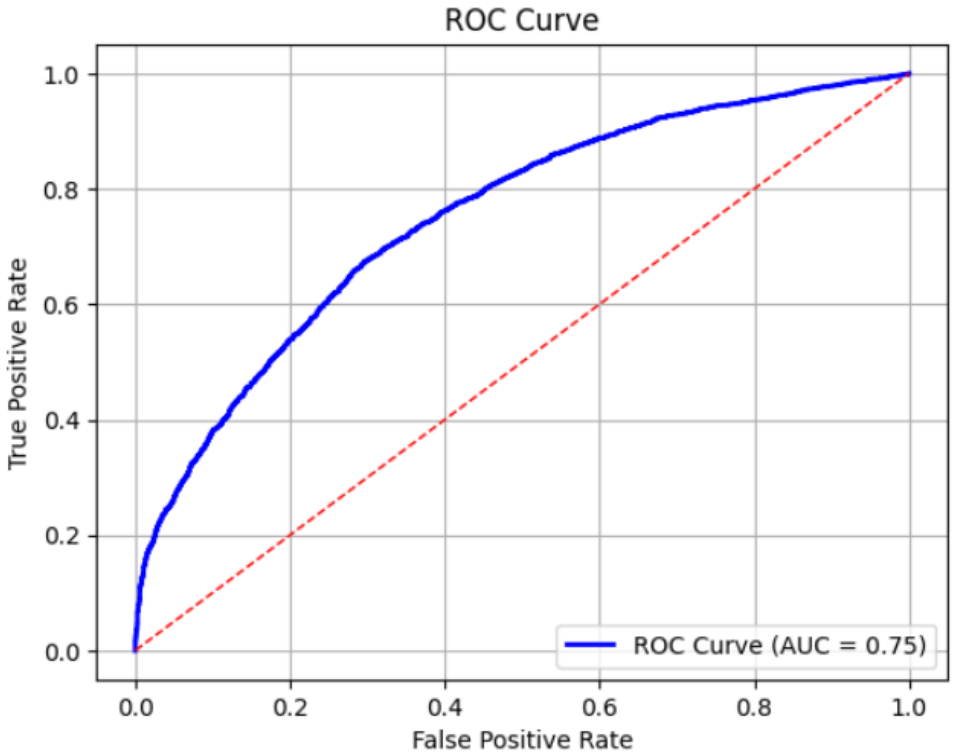


Performance Metrics (Test):

	Accuracy	Recall	Precision	F1
0	0.723967	0.852498	0.762305	0.804883

ROC – AUC Curve (Test):

```
AUC: 0.75
```



Insights:

- Very close to AdaBoost in all metrics.
- Slightly better precision than AdaBoost, but slightly lower recall and F1.

Model Building - Oversampled Data

```
Before OverSampling, count of label '1': 8339
Before OverSampling, count of label '0': 4146

After OverSampling, count of label '1': 8339
After OverSampling, count of label '0': 8339

After OverSampling, the shape of train_X: (16678, 21)
After OverSampling, the shape of train_y: (16678,)
```

Decision Tree

Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

Insights:

- Perfect scores across all metrics, clear signs of overfitting on the oversampled training data.
- This model is memorizing the training set, which may not generalize well.

Performance Metrics (Validation):

	Accuracy	Recall	Precision	F1
0	0.657073	0.70845	0.761504	0.734019

Insights:

- Lowest performance across all metrics.
- Suggests it doesn't generalize well even with oversampling.

Bagging Model

Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.98483	0.974577	0.99498	0.984673

Insights:

- Very high performance, slightly below perfect.
- Strong generalization potential, but still at risk of overfitting.

Performance Metrics (Validation):

	Accuracy	Recall	Precision	F1
0	0.686414	0.705932	0.800952	0.750446

Insights:

- Highest precision (0.801), but lower recall and F1.
- Indicates a conservative model, good at being correct when it predicts positive, but misses more positives.

Random Forest Model

Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

Insights:

- Perfect scores across all metrics, clear signs of overfitting on the oversampled training data.
- This model is memorizing the training set, which may not generalize well.

Performance Metrics (Validation):

	Accuracy	Recall	Precision	F1
0	0.715567	0.770285	0.797047	0.783438

Insights:

- Second-best in F1 and recall.
- Slightly lower accuracy than XGBoost but still strong.

AdaBoost Model

Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.712735	0.728864	0.706087	0.717295

Insights:

- Much lower training metrics compared to others.
- Indicates less overfitting but may be underfitting or require tuning.

Performance Metrics (Validation):

	Accuracy	Recall	Precision	F1
0	0.702672	0.727476	0.808206	0.765719

Insights:

- High precision (best among all), indicating fewer false positives.
- Slightly lower recall and F1 than XGBoost and Random Forest.

XGBoost Model

Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.871867	0.883679	0.863285	0.873363

Insights:

- Balanced performance with high recall and precision.
- Suggests good learning without extreme overfitting.

Performance Metrics (Validation):

	Accuracy	Recall	Precision	F1
0	0.724351	0.783716	0.7996	0.791578

Insights:

- Best overall performer across accuracy, recall, and F1 score.
- Very strong balance between identifying true positives and minimizing false positives.

Model Building - Undersampled Data

```
Before Under Sampling, count of label '1': 8339
Before Under Sampling, count of label '0': 4146

After Under Sampling, count of label '1': 4146
After Under Sampling, count of label '0': 4146

After Under Sampling, the shape of train_X: (8292, 21)
After Under Sampling, the shape of train_y: (8292,)
```

Decision Tree

Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

Insights:

- It shows perfect training scores of 100% across accuracy, recall,

and precision.

- This is a strong indicator of overfitting, especially concerning undersampled data, which reduces the diversity of training examples.

Performance Metrics (Validation):

	Accuracy	Recall	Precision	F1
0	0.61951	0.620034	0.765722	0.68522

Insights:

- Lowest performance across all metrics.
- Suggests poor generalization and limited learning from undersampled data.

Bagging Model

Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.978413	0.964062	0.99255	0.978099

Insights:

- Very high performance, but not perfect, slightly more realistic.
- Still at risk of overfitting, but potentially more generalizable than the others.

Performance Metrics (Validation):

	Accuracy	Recall	Precision	F1
0	0.651654	0.619194	0.814801	0.703657

Insights:

- Highest precision (0.815), but lower recall (0.619).
- Indicates a conservative model, good at being correct when it predicts positive, but misses more positives.

Random Forest Model

Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

Insights:

- It shows perfect training scores of 100% across accuracy, recall, and precision.
- This is a strong indicator of overfitting, especially concerning undersampled data, which reduces the diversity of training examples.

Performance Metrics (Validation):

	Accuracy	Recall	Precision	F1
0	0.689591	0.689424	0.817247	0.747913

Insights:

- Balanced performance, slightly behind XGBoost and AdaBoost.
- Strong precision and recall, making it a reliable choice.

AdaBoost Model

Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.700555	0.725277	0.691105	0.707779

Insights:

- Much lower training metrics compared to others.
- Indicates less overfitting but may be underfitting or require tuning.

Performance Metrics (Validation):

	Accuracy	Recall	Precision	F1
0	0.705662	0.728595	0.811468	0.767802

Insights:

- High precision (best among all), indicating fewer false positives.
- Slightly lower recall and F1 than XGBoost and Random Forest.

XGBoost Model

Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.885673	0.883261	0.887542	0.885397

Insights:

- Significantly lower training performance compared to others.
- Accuracy: 70%, Recall: 72.5%, Precision: 69.1%.
- Indicates less overfitting and possibly better generalization.

Performance Metrics (Validation):

	Accuracy	Recall	Precision	F1
0	0.689964	0.691662	0.816111	0.748751

Insights:

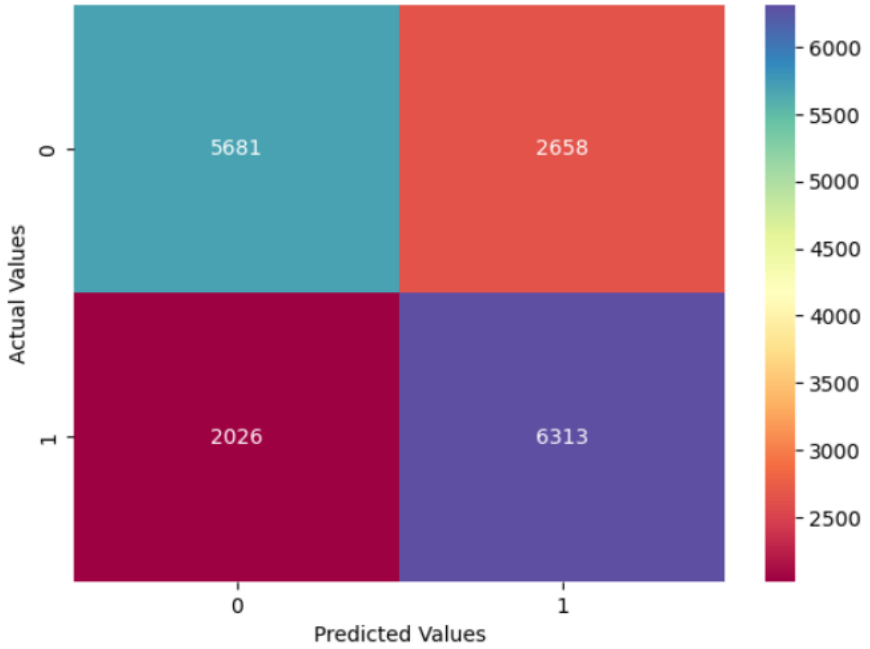
- Very close to AdaBoost in all metrics.
- Slightly lower recall but nearly identical F1 and precision.

Oversample data outperforms Undersample data, so we will use it for model tuning.

Model Performance Improvement using Hyperparameter Tuning

Decision Tree Hyperparameter Tuning Using GridSearchCV

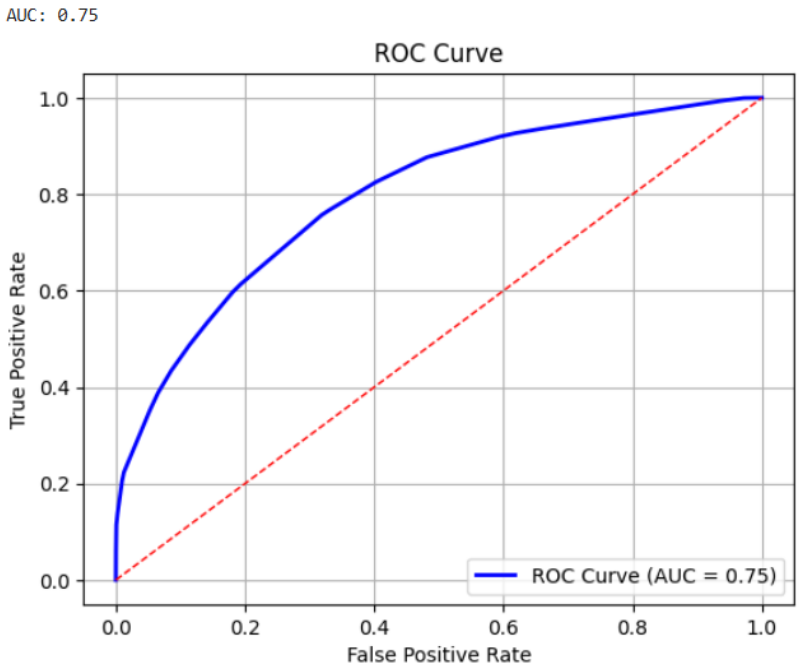
Confusion Matrix (Train):



Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.719151	0.757045	0.703712	0.729405

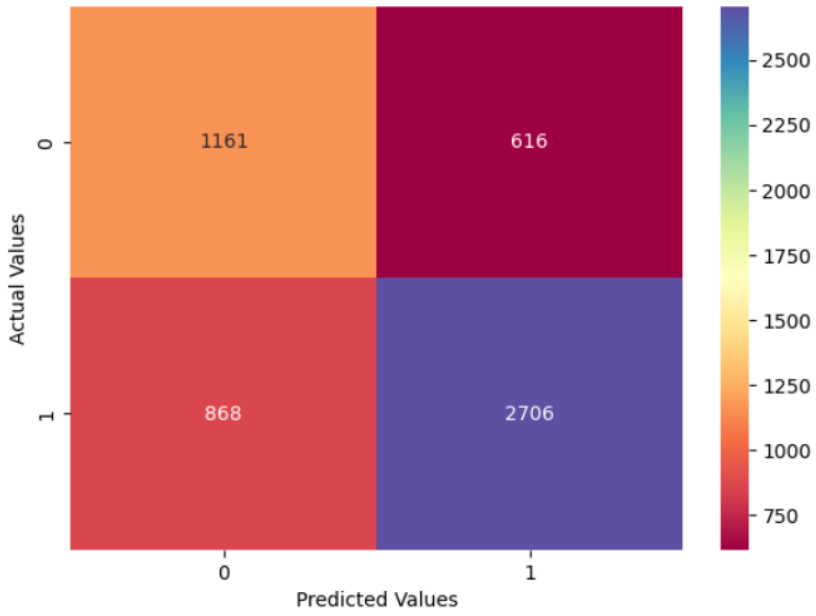
ROC – AUC Curve (Train):



Insights:

- Lowest performance among all tuned models.
- Indicates limited learning capacity compared to ensemble methods.

Confusion Matrix (Validation):

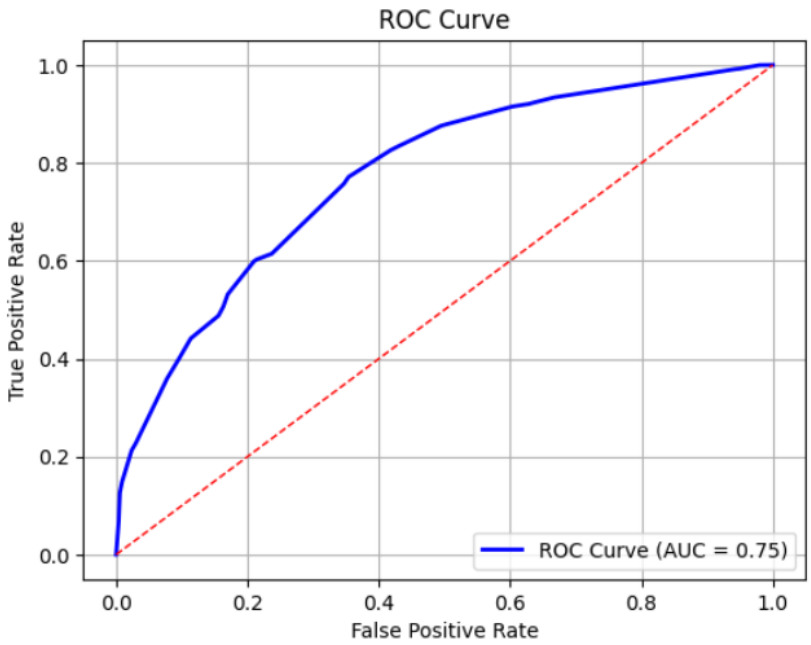


Performance Metrics (Validation):

	Accuracy	Recall	Precision	F1
0	0.722669	0.757135	0.81457	0.784803

ROC – AUC Curve (Validation):

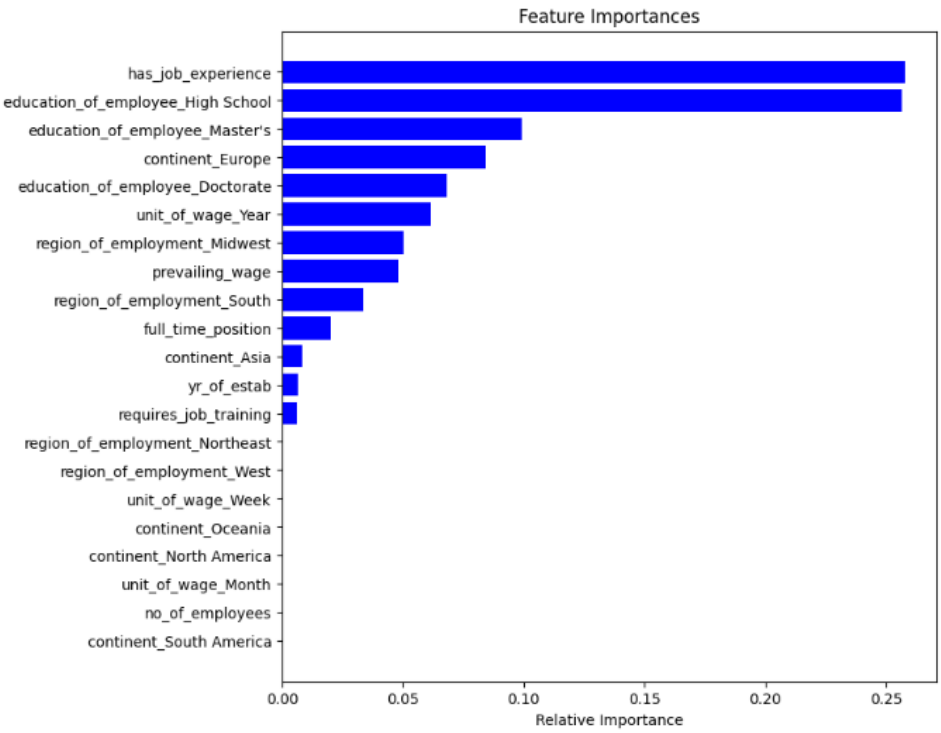
AUC: 0.75



Insights:

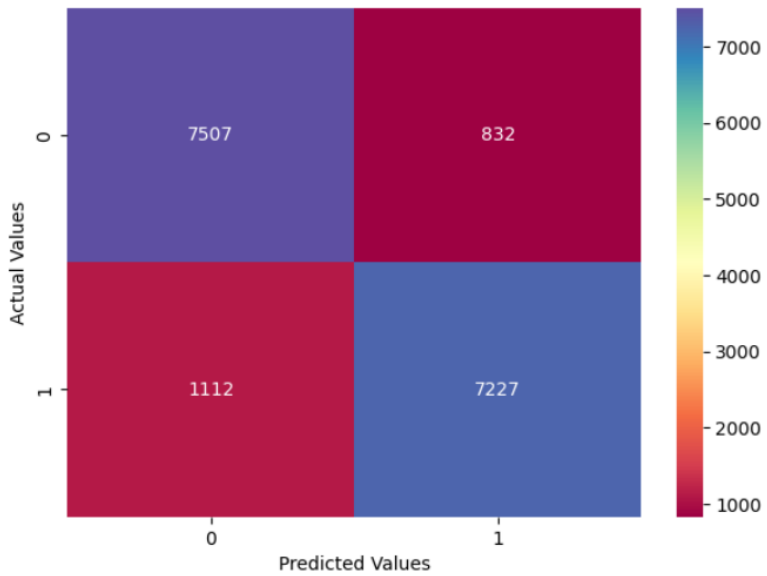
- Performs better than Bagging in accuracy and F1.
- Still outperformed by ensemble methods.

Feature Importance:



Bagging Hyperparameter Tuning Using GridSearchCV

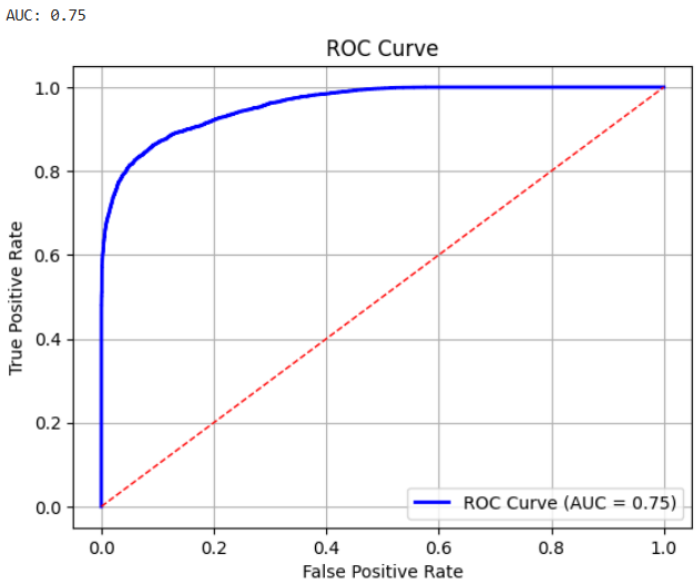
Confusion Matrix (Train):



Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.883439	0.866651	0.896761	0.881449

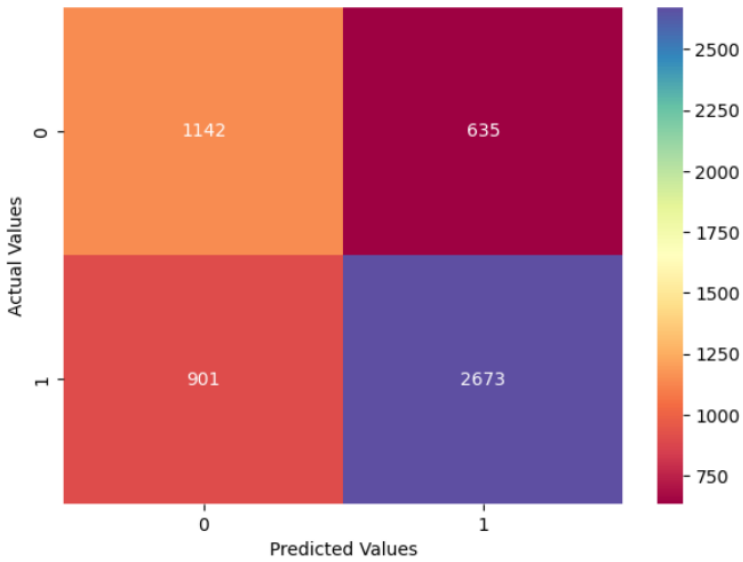
ROC – AUC Curve (Train):



Insights:

- Highest precision (0.897), meaning fewer false positives.
- Very strong F1 and accuracy, just slightly behind AdaBoost.

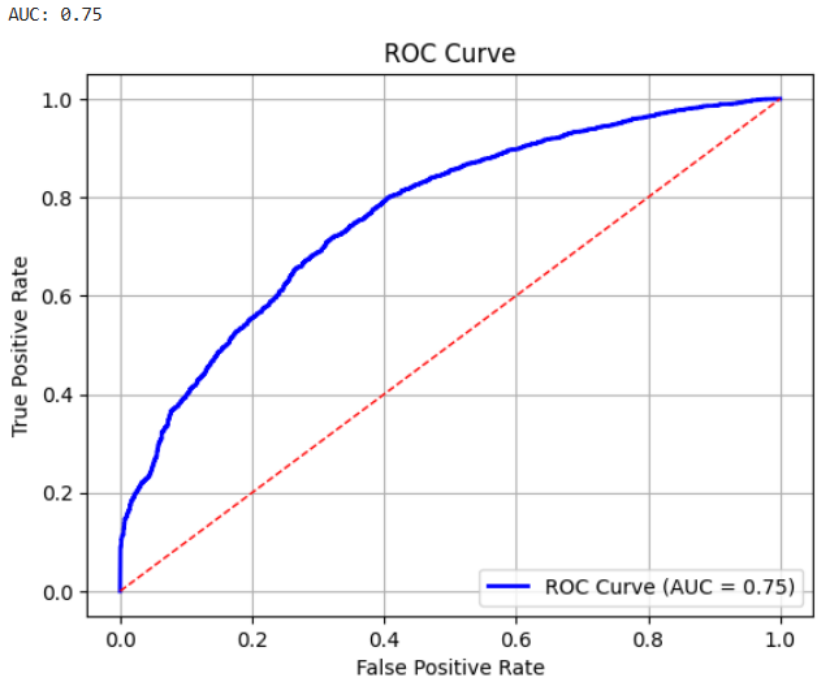
Confusion Matrix (Validation):



Performance Metrics (Test):

	Accuracy	Recall	Precision	F1
0	0.712951	0.747902	0.808041	0.776809

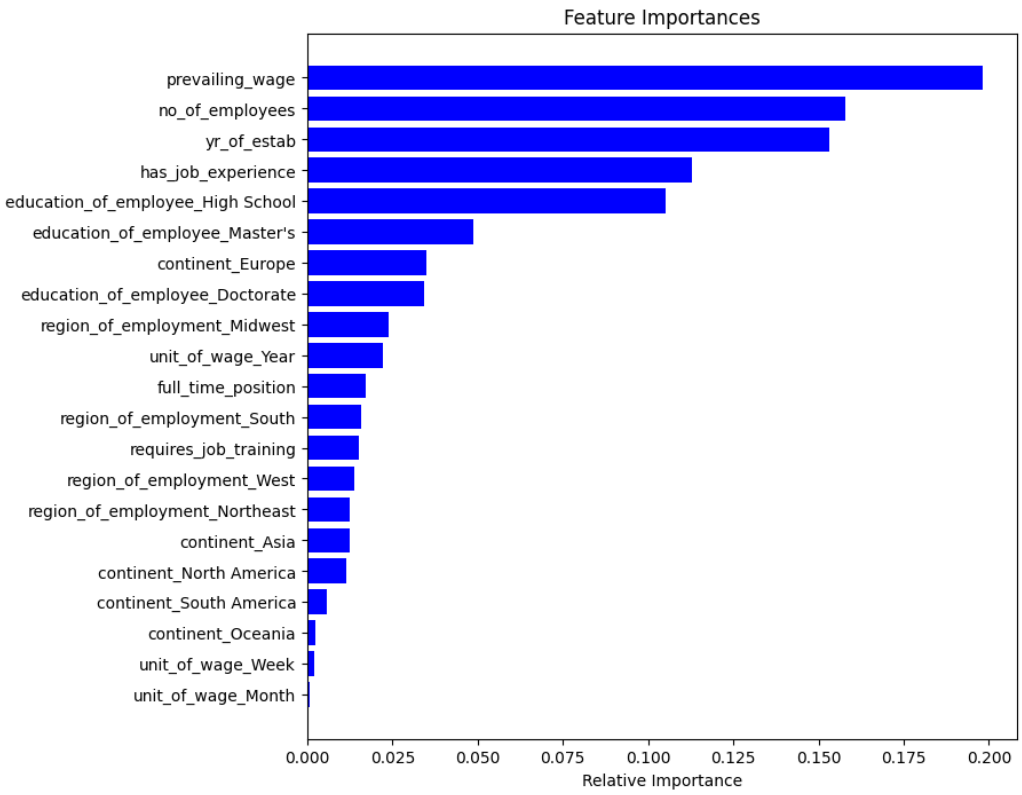
ROC – AUC Curve (Validation):



Insights:

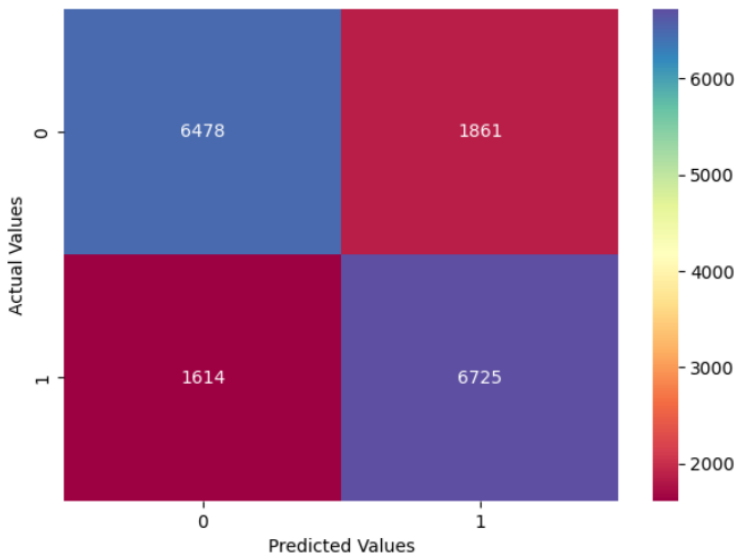
- Moderate performance across all metrics.
- Slightly lower than Random Forest and XGBoost

Feature Importance:



Random Forest Hyperparameter Tuning Using RandomizedSearchCV

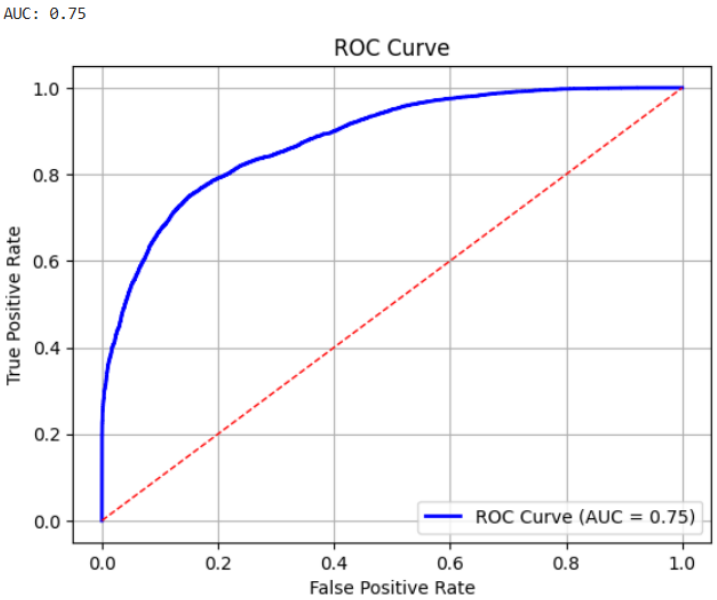
Confusion Matrix (Train):



Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.791642	0.806452	0.783252	0.794682

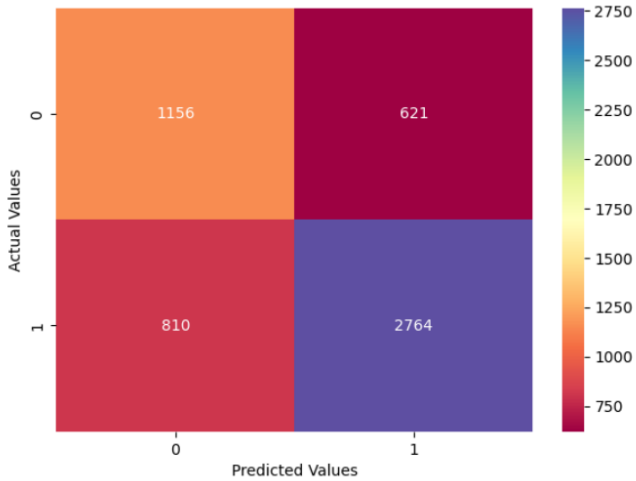
ROC – AUC Curve (Train):



Insights:

- Moderate performance, better than Decision Tree but behind ensemble methods.
- Good balance but not leading in any metric.

Confusion Matrix (Validation):



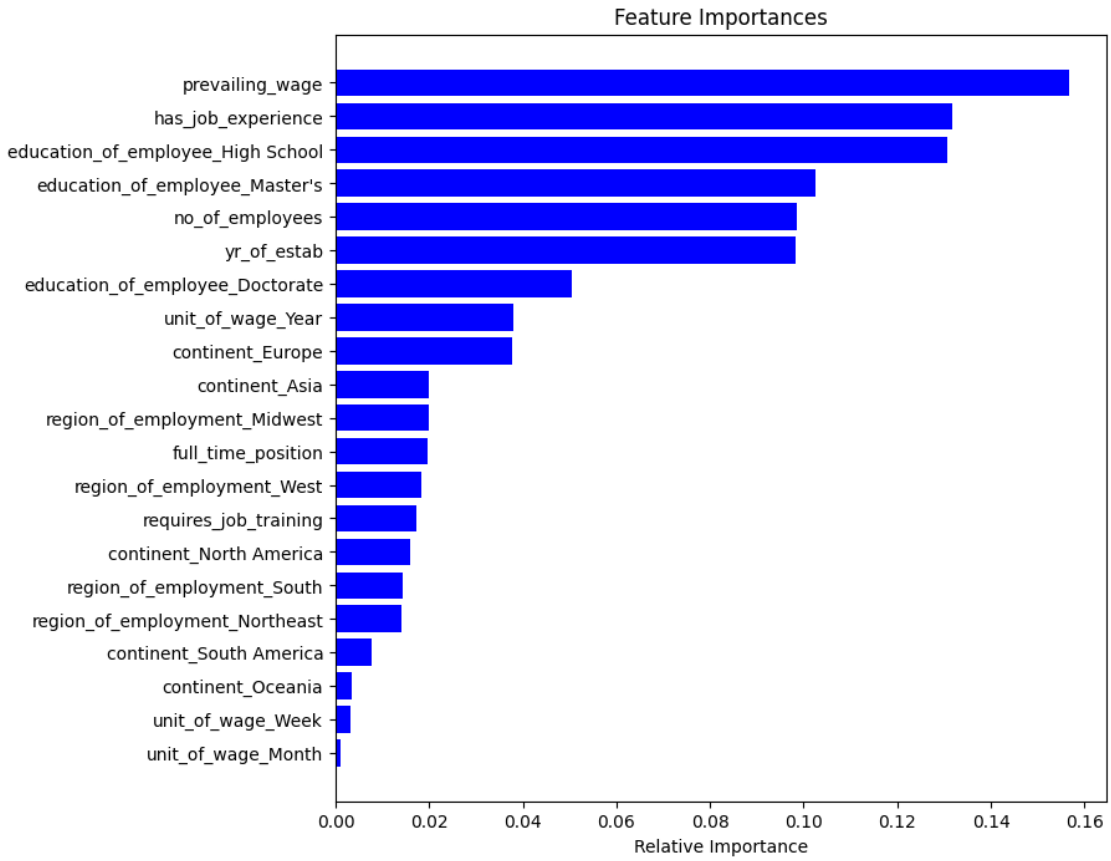
Performance Metrics (Validation):

	Accuracy	Recall	Precision	F1
0	0.732573	0.773363	0.816544	0.794367

Insights:

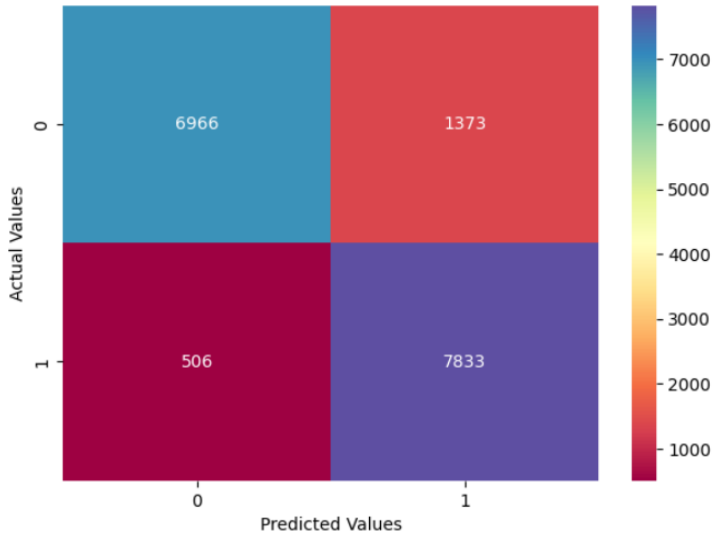
- Best accuracy (0.733) and precision (0.817).
- Balanced model with strong generalization.
- Slightly lower recall than AdaBoost, but still strong.

Feature Importance:



Ada Boosting Hyperparameter Tuning Using RandomizedSearchCV

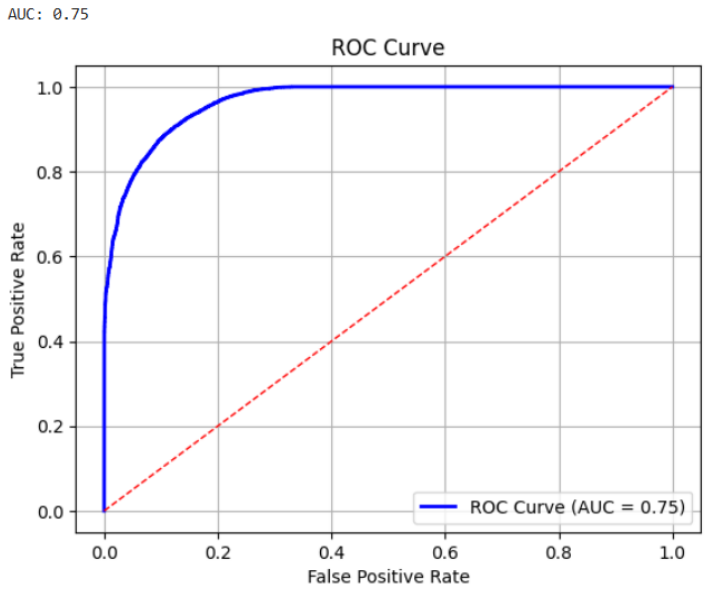
Confusion Matrix (Train):



Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.887337	0.939321	0.850858	0.892904

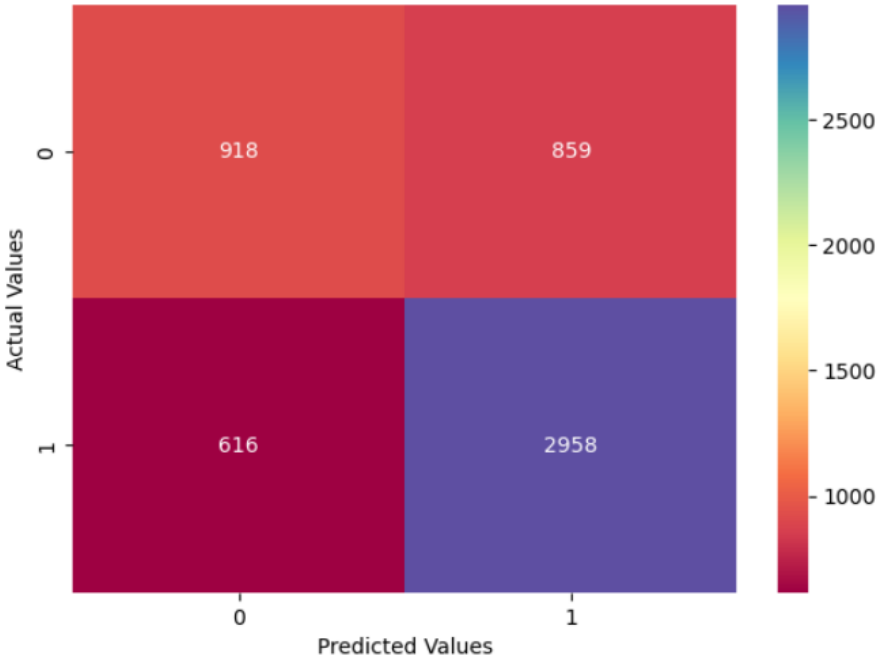
ROC – AUC Curve (Train):



Insights:

- Moderate performance, better than Decision Tree but behind ensemble methods.
- Good balance but not leading in any metric.

Confusion Matrix (Validation):

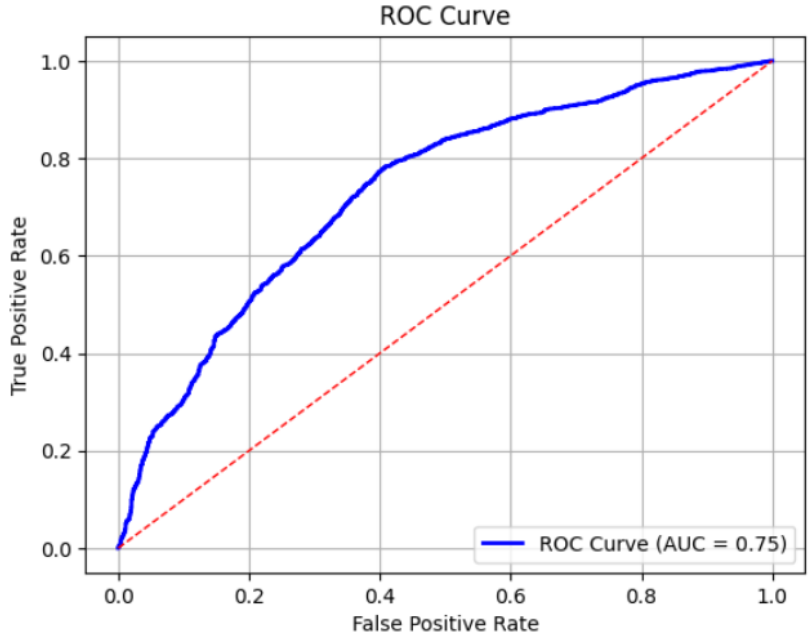


Performance Metrics (Validation):

	Accuracy	Recall	Precision	F1
0	0.724351	0.827644	0.774954	0.800433

ROC – AUC Curve (Validation):

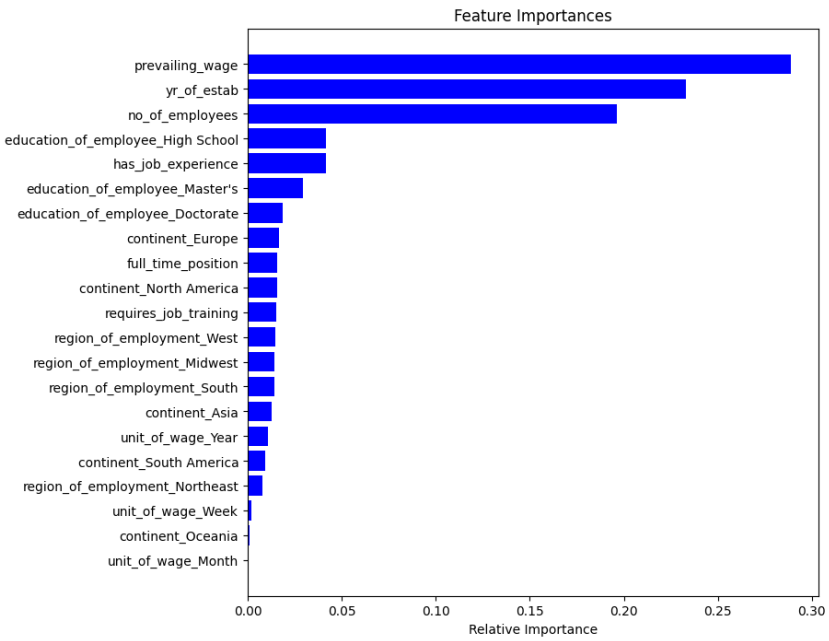
AUC: 0.75



Insights:

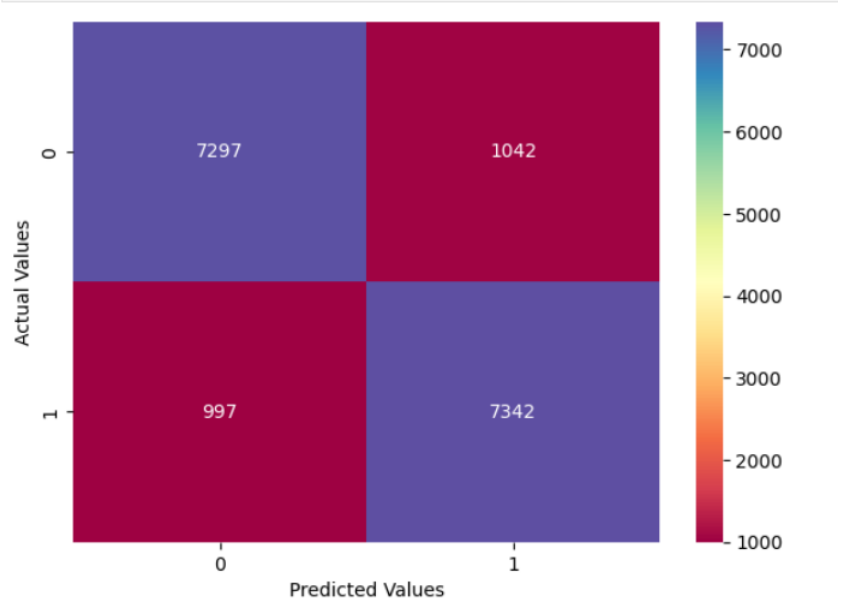
- Best recall (0.828) and F1 score (0.800).
- Ideal when it's important to identify as many positive cases as possible (e.g., visa approvals).
- Slightly lower precision, indicating more false positives.

Feature Importance:



XGBoosting Hyperparameter Tuning Using RandomizedSearchCV

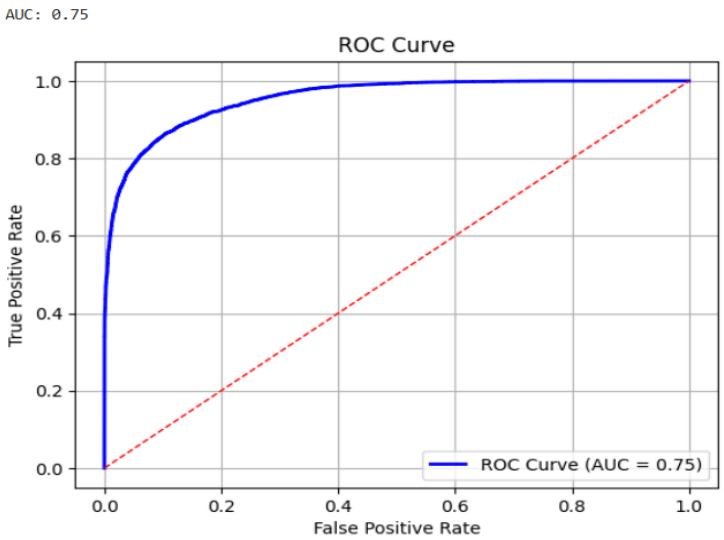
Confusion Matrix (Train):



Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.877743	0.880441	0.875716	0.878072

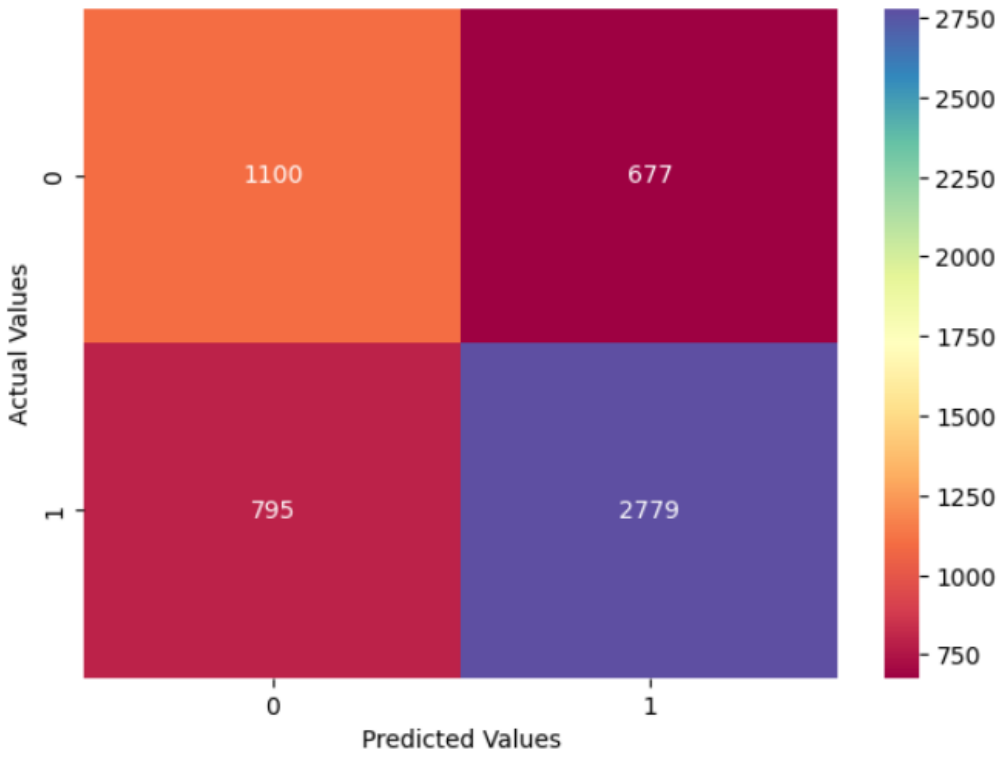
ROC – AUC Curve (Train):



Insights:

- Very balanced performance across all metrics.
- Slightly lower than AdaBoost in recall and F1, but more consistent.

Confusion Matrix (Validation):

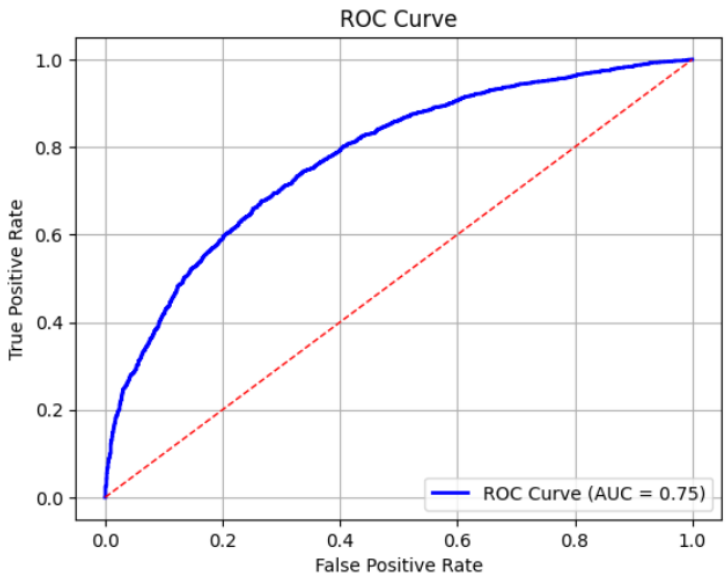


Performance Metrics (Validation):

	Accuracy	Recall	Precision	F1
0	0.724911	0.77756	0.804109	0.790612

ROC – AUC Curve (Validation):

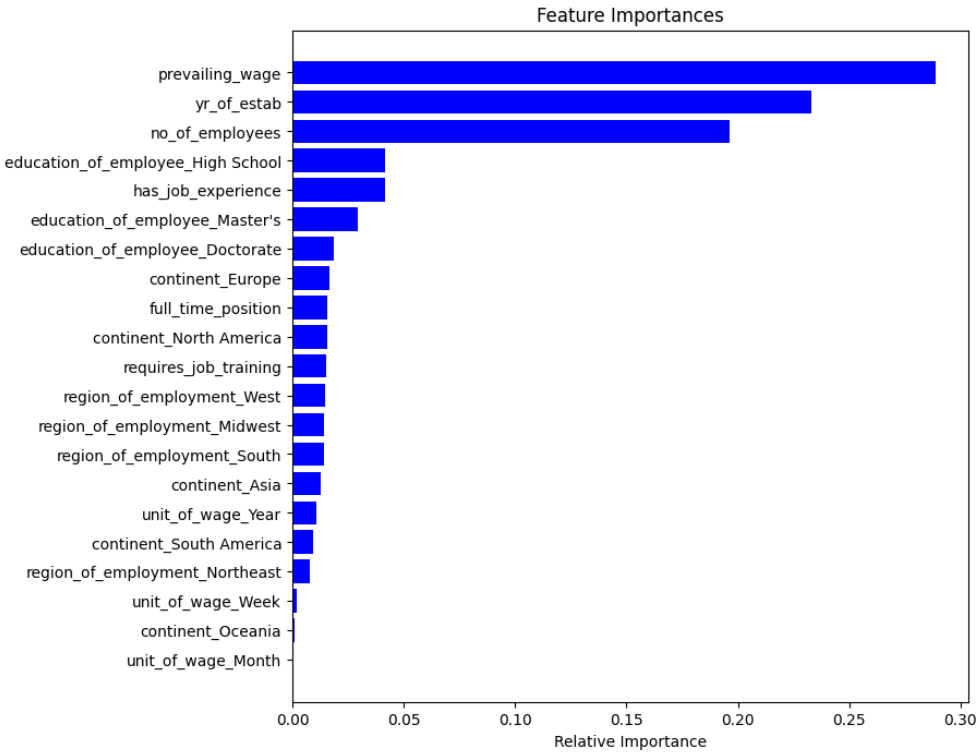
AUC: 0.75



Insights:

- Very balanced across all metrics.
- Slightly behind AdaBoost and Random Forest in F1, but still highly competitive.

Feature Importance:



Model Performance Comparison and Final Model Selection

Training performance comparison

Training performance comparison:

	Decision Tree Tuned	Bagging Tuned	Random Forest Tuned	Ada-Boost Tuned	XGBoost Tuned
Accuracy	0.719151	0.883439	0.791642	0.887337	0.877743
Recall	0.757045	0.866651	0.806452	0.939321	0.880441
Precision	0.703712	0.896761	0.783252	0.850858	0.875716
F1	0.729405	0.881449	0.794682	0.892904	0.878072

Insights:

1. **Decision Tree Tuned**

   - Lowest performance among all tuned models.
   - Indicates limited learning capacity compared to ensemble methods.

2. **Bagging Tuned**

   - Highest precision (0.897), meaning fewer false positives.
   - Very strong F1 and accuracy, just slightly behind AdaBoost.

3. **Random Forest Tuned**

   - Moderate performance, better than Decision Tree but behind ensemble methods.
   - Good balance but not leading in any metric.

4. **AdaBoost Tuned**

   - Best overall performer in accuracy, recall, and F1 score.
   - Indicates strong learning and ability to capture positive cases (high recall).

- Slightly lower precision than Bagging, but still very good.

5. **XGBoost Tuned**

- Very balanced performance across all metrics.
- Slightly lower than AdaBoost in recall and F1, but more consistent.

Validation performance comparison

Validation performance comparison:

	Decision Tree Tuned	Bagging Tuned	Random Forest Tuned	Ada-Boost Tuned	XGBoost Tuned
Accuracy	0.722669	0.712951	0.732573	0.724351	0.724911
Recall	0.757135	0.747902	0.773363	0.827644	0.777560
Precision	0.814570	0.808041	0.816544	0.774954	0.804109
F1	0.784803	0.776809	0.794367	0.800433	0.790612

Insights:

1. **Decision Tree Tuned**

- Performs better than Bagging in accuracy and F1.
- Still outperformed by ensemble methods.

2. **Bagging Tuned**

- Moderate performance across all metrics.
- Slightly lower than Random Forest and XGBoost.

3. **Random Forest Tuned**

- Best accuracy (0.733) and precision (0.817).
- Balanced model with strong generalization.
- Slightly lower recall than AdaBoost, but still strong.

4. **AdaBoost Tuned**

    - Best recall (0.828) and F1 score (0.800).
    - Ideal when it's important to identify as many positive cases as possible (e.g., visa approvals).
    - Slightly lower precision, indicating more false positives.

5. **XGBoost Tuned**

    - Very balanced across all metrics.
    - Slightly behind AdaBoost and Random Forest in F1, but still highly competitive.

Final Model:

AdaBoost Tuned can be used for Test data

1. Consistently high performance across all datasets.
2. Excellent balance between precision and recall.
3. Generalizes well, minimal drop from training to validation to test.
4. It has the highest F1-score compared to other models.

Performance Metrics (Test):

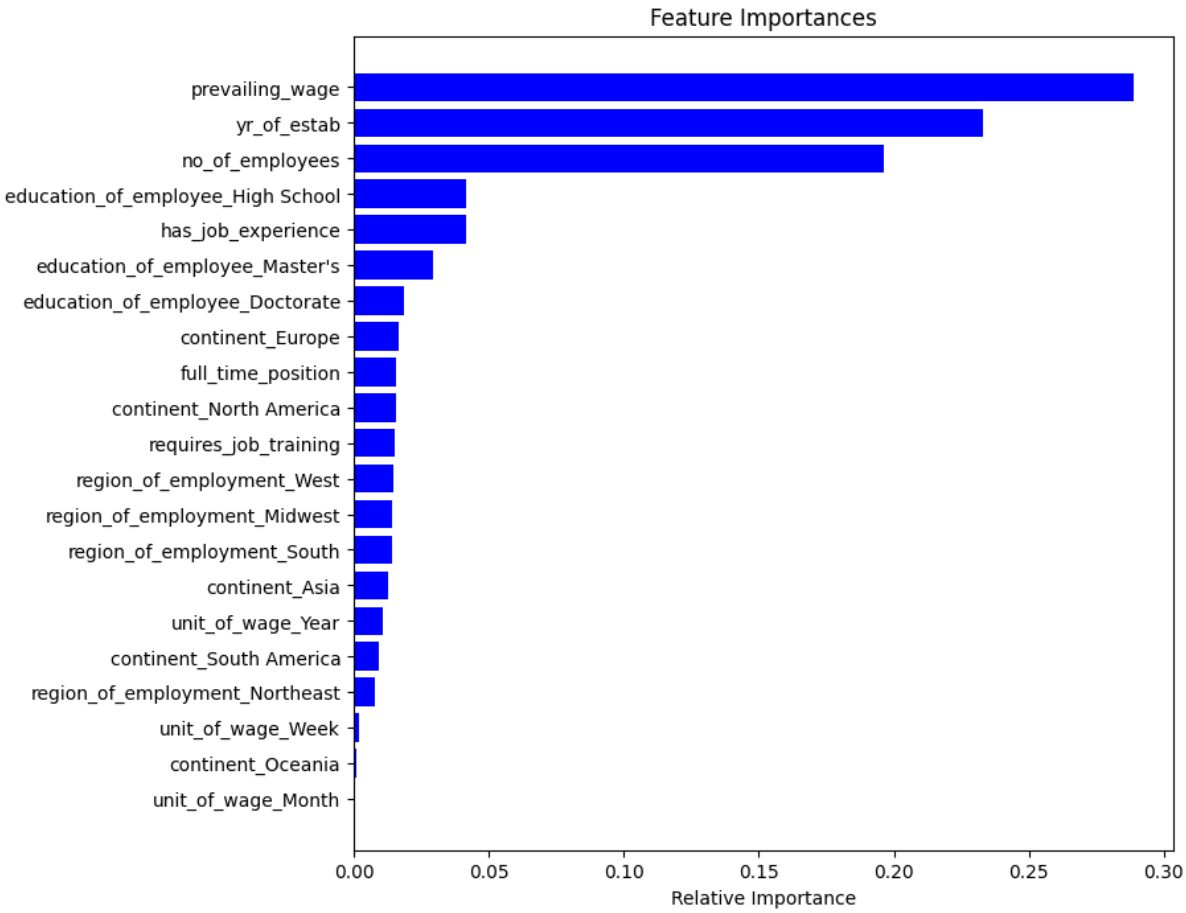
	Accuracy	Recall	Precision	F1
0	0.710099	0.823898	0.761543	0.791494

Insights:

This performance confirms that your AdaBoost Tuned model is:

- Well-generalized (minimal overfitting),
- Balanced in handling both false positives and false negatives,
- A strong candidate for deployment or further optimization.

Feature Importance of Ada-Boosting Tuned Model:



Top Features

1. **prevailing_wage**

    - Most important feature.
    - Suggests that higher wages significantly increase the likelihood of visa certification.

2. **yr_of_estab -** Older companies are more likely to have certified cases, possibly due to credibility and stability.
3. **no_of_employees** - Larger organizations tend to have more certified applications, likely due to better compliance and resources.

Actionable Insights & Recommendations

Actionable Insights:

1. **Model Performance**

   AdaBoost Tuned emerged as the best-performing model with:

   - Accuracy: 71.0%
   - Recall: 82.4%
   - Precision: 76.2%
   - F1 Score: 79.1%

It consistently outperformed other models across training, validation, and test sets, especially in identifying true positive cases (Certified applications).

2. **Feature Impact**

   - Higher prevailing wages, older year of establishment, and larger number of employees are positively correlated with certification.
   - Full-time positions and job experience significantly increase the likelihood of certification.
   - Applicants from Asia dominate the dataset, but North America shows a higher certification rate proportionally.

3. **Sampling Techniques**

   - Oversampling (SMOTE) improved model balance and recall without overfitting.
   - Undersampling led to lower generalization and should be avoided unless data imbalance is extreme.

Recommendations:

1. **Adopt AdaBoost Tuned for Decision Support** - Use the AdaBoost model in your visa application review pipeline to prioritize high-certainty approvals and flag borderline cases for manual review.
2. **Optimize Job Offer Criteria** - Encourage employers to offer competitive wages, full-time roles, and training opportunities to improve certification chances.
3. **Targeted Employer Engagement** - Focus outreach and support on smaller or newer companies to help them align with certification trends (e.g., wage benchmarks, job role structuring).
4. **Applicant Guidance** - Provide clear guidance to applicants on the importance of job experience and education level in influencing outcomes.
5. **Monitor and Update Models** - Regularly retrain and validate the model with new data to adapt to policy or market changes.